

Adventist University of Central Africa

PROTRACK AI: INTELLIGENT EMPLOYEE PRODUCTIVITY
MANAGEMENT SYSTEM
CASE STUDY: MTN RWANDA

A Final year Project Presented in Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Information Technology

Major in Software Engineering

By

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ABSTRACT

A final year Project for the Bachelor Degree in Information Technology

Emphasis in Software Engineering

Adventist University of Central Africa

TITLE: PROTRACK AI: INTELLIGENT EMPLOYEE PRODUCTIVITY MANAGEMENT SYSTEM

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Employee productivity management is a growing challenge in digital workplaces. At MTN Rwanda, a two-month internship revealed that the existing monitoring system operated purely as a passive internet logger, unable to distinguish productive from non-productive digital activity. Employees freely accessed social media and video platforms during work hours with no management intervention, resulting in significant time wastage and suboptimal performance.

Further investigation confirmed that while the system technically possessed website-blocking capabilities, these were never deployed. Interviews with colleagues and a supervisor established that the absence of productivity oversight was standard practice, not a temporary gap. Performance evaluations relied entirely on subjective observations and output-based metrics, with no analysis of actual digital work behavior, demonstrating a critical need for an intelligent productivity management solution. ProTrackAI was developed to address these gaps, an intelligent system that monitors, classifies, and optimizes workforce digital engagement in real time.

Data was collected through direct observation and informal interviews, and the Agile methodology guided development through iterative sprints. Systems analysis and design utilized object oriented diagrams to define objectives. Tools used: Frontend: React.js and Material UI; Backend: Node.js, Express.js, scikit-learn and TensorFlow.js; Database: MongoDB.

Key features include automated activity classification, productivity dashboards, employee self-evaluation portals, and goal management tools, enabling organizations to reduce time wastage, optimize resource allocation, and support informed HR decisions in Rwanda's growing digital economy.

DECLARATION

I, Jesse Shema Kyanga, Registration Number 25147, a student at the Adventist University of Central Africa in the Faculty of Information Technology, Department of Software Engineering, solemnly declare that this final year project titled "PROTRACKAI - INTELLIGENT EMPLOYEE PRODUCTIVITY MANAGEMENT SYSTEM" is my original work, conducted under the supervision of UZAYISENGA Emmanuel. To the best of my knowledge, it reflects my own research, findings, and contributions, and all external sources have been duly acknowledged.

Signature:

Date:/...../.....

APPROVAL

I, UZAYISENGA Emmanuel, hereby certify that this final year project report has been done under my supervision and submitted with my approval.

Signature:

Date:/...../.....

DEDICATION

To the Almighty, whose blessings and grace guided me throughout this entire academic journey.

To my parents and family, whose sacrifices and prayers built the foundation of my success.

To my supervisor, UZAYISENGA Emmanuel, whose mentorship sharpened my research.

To my siblings, whose encouragement and support kept me going.

To my friends, whose camaraderie enriched this journey.

To my classmates, who made this worthwhile.

To everyone who believed in me.

To those who inspired me.

To God, always.

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LIST OF ABBREVIATIONS

AUCA:	Adventist University of Central Africa
AI:	Artificial Intelligence
API:	Application Programming Interface
CORS:	Cross-Origin Resource Sharing
CPU:	Central Processing Unit
CRUD:	Create, Read, Update, Delete
CSS:	Cascading Style Sheet
HTML:	Hypertext Markup Language
HTTP:	Hypertext Transfer Protocol
IDE:	Integrated Development Environment
JSON:	JavaScript Object Notation
JWT:	JSON Web Token
KPI:	Key Performance Indicator
ML:	Machine Learning
NoSQL:	Not Only Structured Query Language
RAM:	Random Access Memory
REST:	Representational State Transfer
UI:	User Interface
URL:	Uniform Resource Locator
UX:	User Experience

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I begin by expressing my profound gratitude to God Almighty, whose continued blessings and guidance have sustained me throughout this academic endeavor. His strength has been my constant source of perseverance.

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My sincere thanks go to the entire Administration and Academic Staff of the Adventist University of Central Africa (AUCA) for providing an enriching learning environment. The institution's commitment to excellence and the dedication of its faculty have been pivotal in shaping my academic journey.

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To my family, whose sacrifices and prayers have been the bedrock of my educational pursuits, I owe more than words can convey. Your belief in my abilities has propelled me forward even in moments of doubt.

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Lastly, to everyone who contributed to this work in ways both large and small, I sincerely thank you for being instrumental in this achievement.

May God Bless

Jesse Shema Kyanga

CHAPTER 1

GENERAL INTRODUCTION

Introduction

The contemporary workplace has undergone a dramatic transformation with the widespread adoption of digital technologies and remote work models(Brynjolfsson et al., 2020) . While these changes have introduced unprecedented flexibility and connectivity, they have simultaneously created new challenges in measuring and maintaining employee productivity. Organizations increasingly struggle to distinguish genuine work engagement from digital distractions, particularly as employees spend the majority of their workday interacting with computers and online platforms.

In Rwanda's rapidly evolving digital economy, companies like MTN Rwanda face mounting pressure to optimize workforce productivity while maintaining service quality standards. The absence of intelligent monitoring systems means managers often rely on subjective assessments rather than objective data when evaluating employee performance. This gap between technological capability and management practice creates opportunities for substantial productivity losses that directly affect organizational competitiveness and profitability.

Traditional approaches to workplace monitoring have proven inadequate for modern digital work environments. Basic network filtering systems can block access to inappropriate websites, but they cannot differentiate between productive and non-productive use of permitted platforms. An employee might spend hours on YouTube watching either work-related tutorials or entertainment content yet current monitoring tools treat these activities identically. This lack of contextual understanding leaves managers without the granular insights needed to support struggling employees or recognize high performers.

This final year project addresses these challenges by developing ProTrackAI, an artificial intelligence-powered productivity management system designed specifically for the Rwandan workplace context. By combining real-time activity monitoring with machine learning classification algorithms, the system provides organizations with the objective data needed to optimize workforce performance while creating transparency and accountability frameworks that benefit both employers and employees.

Background of the Study

The rapid development of information and communication technologies has significantly transformed the modern workplace, shifting organizational operations from physical tasks to digital platforms. While this transformation has improved global collaboration and efficiency, it has introduced significant challenges regarding the monitoring of employee productivity. In traditional work environments, managers assessed performance through direct supervision and observable physical presence. However, as work increasingly occurs within digital systems, determining how employees utilize their time has become complex, necessitating the integration of technological monitoring systems to maintain organizational oversight and operational transparency.

The relationship between information technology and productivity remains a central theme in academic research. This discourse began with the "productivity paradox," where initial large-scale investments in information technology did not immediately result in measurable productivity gains (Brynjolfsson, 1993). Subsequent research clarified that these gains are not automatic; rather, they depend on how effectively technology is integrated into specific management practices and organizational processes (Brynjolfsson & Yang, 1996). Consequently, modern organizations have moved toward digital analytics to track computer usage and online activity. As noted by Lehmann and Beckmann (2024), these monitoring technologies reduce the cost of performance evaluation by providing continuous, objective data that replaces subjective observation.

To address these digital shifts, many organizations have adopted monitoring and analytics systems to track computer usage and online activity. These systems can collect granular information such as website visits, application usage, and system login times. As workplaces continue to become more digitized, organizations are increasingly exploring the use of data analytics and artificial intelligence to better understand employee behavior and improve productivity management. In addition, data-driven productivity management has become an important component of modern organizational strategy. Studies show that organizations that integrate data analytics into management practices are better able to support employee productivity (McAfee & Brynjolfsson, 2012).

In the telecommunications sector, this reliance on digital systems is absolute. Within large organizations such as MTN Rwanda, employees depend on corporate networks for customer support, network monitoring, communication, and administrative operations. However, this environment presents a unique challenge: employees have constant access to various websites and applications that may not always be directly related to their assigned duties. Without an effective monitoring system, it becomes difficult for management to determine whether employees are using their working hours productively or engaging in non-work-related activities. Furthermore, existing records such as simple login times or network usage logs often lack the analytical depth required to identify long-term productivity trends, leaving a gap between data collection and actionable insight.

To address the limitations of traditional monitoring, this study proposes ProTrackAI, an intelligent productivity management system designed to categorize and analyze digital activity. Unlike passive logging systems, ProTrackAI utilizes machine learning techniques to classify website visits and application usage into productive and non-productive categories based on predefined behavioral patterns. This transition from manual observation to automated analysis aligns with modern organizational trends where artificial intelligence is used to automate analytical tasks and support complex decision-making processes (Davenport & Ronanki, 2018). The proposed system collects information related to employee computer usage and generates analytical reports that allow managers to identify work patterns and potential inefficiencies.

By integrating intelligent data analysis with real-time monitoring, ProTrackAI provides managers with data-driven indicators of workforce performance. This approach allows for a more nuanced understanding of how digital resources are utilized, supporting the identification of workflow bottlenecks. Ultimately, organizations that integrate such data analytics into their management strategies are better positioned to enhance employee productivity and maintain a competitive edge in an increasingly digitized economy. The implementation of ProTrackAI represents a move toward a more intelligent, objective, and scalable solution for managing employee output in the modern Rwandan telecommunications landscape.

Statement of the Problem

MTN Rwanda currently operates with a monitoring system that fails to address employee productivity management in digital work environments. The existing portal-based internet access system functions solely as a content filter, blocking inappropriate websites while providing no meaningful insights into how employees utilize their work time.

The system cannot distinguish between productive and non-productive use of permitted websites. An employee spending hours on YouTube watching entertainment is treated identically to one researching technical solutions. This lack of contextual understanding means substantial paid work time may be consumed by non-work activities without detection. The reliance on a single staff member to oversee internet usage across the organization is neither scalable nor effective. With no active monitoring or automated alerts, productivity patterns go untracked and time-wasting behaviors remain unidentified. Managers lack objective data to inform performance evaluations, identify training needs, or recognize high performers.

Inability to Differentiate Website Usage: The system cannot distinguish between productive and non-productive use of permitted websites, treating entertainment activities like YouTube the same as work-related research, which may lead to lost work time.

Single-Person Monitoring Limitation: Reliance on a single staff member to oversee internet usage is not scalable or effective, leaving many activities unmonitored.

Lack of Automated Tracking and Alerts: Without real-time monitoring or automated notifications, time-wasting behaviors go undetected, and productivity patterns remain unclear.

Limited Data for Management Decisions: Managers do not have objective data to evaluate performance, identify training needs, or recognize high-performing employees, reducing the ability to improve overall productivity.

No Historical Usage Analysis: The system does not maintain records of past internet activity, making it difficult to identify recurring issues or trends over time.

Inefficient Performance Accountability: Employees' online activities cannot be correlated with work output, limiting accountability and performance management, leading to the inability to know the true productivity of a certain employee or a team or a department, leading to inefficient employee management.

Choice and Motivation of the Study

This study was motivated by the growing need for intelligent productivity management solutions in digital workplaces. The decision to develop ProTrackAI stemmed from direct observation of productivity challenges at MTN Rwanda, combined with the opportunity to apply artificial intelligence to address a real and pressing organizational problem.

To AUCA: This final year project to Auca's mission to produce graduates who develop innovative solutions for African workplace challenges. It demonstrates how academic training in the IT Faculty, Department of Software Engineering can translate into meaningful technological contributions that address real industry needs. ProTrackAI exemplifies the practical application of software engineering knowledge to enhance organizational efficiency within Rwanda's growing digital economy.

To (MTN Rwanda): ProTrackAI directly addresses the productivity monitoring gaps observed at MTN Rwanda. By providing management with objective, data-driven insights into employee digital activity, the system enables informed decision-making around performance evaluations, resource allocation, and training needs. Rather than relying on subjective assessments, MTN Rwanda's administrators and faculty staff would gain the visibility needed to optimize workforce productivity while maintaining employee privacy and autonomy.

To the Student: This project represents a personal opportunity to apply software engineering training to solve a real problem encountered during a professional internship. Witnessing firsthand how the absence of intelligent monitoring affected workplace productivity inspired the development of a system that could create meaningful organizational change. ProTrackAI reflects the practical culmination of academic learning applied to authentic workplace challenges. Furthermore, it strengthens professional competence by showcasing the ability to analyze organizational needs, design intelligent solutions, and deliver a system aligned with real operational requirements. In addition, the project builds confidence and readiness to tackle complex technological problems that may be encountered while in work environments like that of MTN Rwanda, this trains the ability to analyze problems and come up with suitable solutions that are maintainable, scalable and reliable.

Objectives of the Study

This final year project aims to develop an intelligent productivity management system that addresses the limitations observed in current monitoring approaches, providing organizations with actionable insights while maintaining employee privacy and fostering accountability.

General Objective

To develop an AI-powered employee productivity monitoring system that accurately tracks and classifies digital workplace activities to improve workforce management and organizational efficiency at MTN Rwanda. The system aims to provide managers with objective, data-driven insights into employee activity patterns and identify areas for performance improvement, while ensuring fair and consistent monitoring across the organization.

Specific Objectives

- To develop a real-time tracking and classification engine: The primary technical step is to create a background agent capable of monitoring employee activity across web browsers and desktop applications. This involves implementing machine learning algorithms to immediately classify digital engagement as productive or non-productive based on job specific roles, which forms the foundational data layer of the entire system.
- To establish a transparent framework for goal setting and progress monitoring: Once the tracking infrastructure is in place, the system will allow managers to define specific professional goals and milestones. This objective focuses on creating the logic that maps real-time digital activity against these predefined targets, ensuring that the software can measure progress toward organizational objectives as they occur.
- To design and implement an employee self-evaluation portal: Following the establishment of goals, the system will provide a user facing interface where employees can view their own real-time analytics. This allows staff to compare their current activity against their set goals and make autonomous adjustments to their work habits before a formal review.
- To engineer a comprehensive reporting and analytics dashboard: The final step involves synthesizing all evaluated data into high level professional reports. This allows the appropriate authorities to view the performance of the department or team.

Scope of the Study

This study focuses on the design, development, and implementation of ProTrackAI, an AI powered employee productivity management system engineered to address digital workplace monitoring challenges at MTN Rwanda. The scope of this research is defined by the technical boundaries of the software and the organizational environment in which it is deployed. The project encompasses several core functional areas, including real-time activity tracking, machine learning based classification, and hierarchical reporting structures.

Technically, the study covers the development of a desktop client and a centralized web dashboard. The system is designed to monitor and classify employee activity across web browsers and desktop applications during official working hours. By utilizing machine learning algorithms, the software distinguishes between productive and non-productive engagement based on specific job roles. Furthermore, the scope includes a comprehensive administrative portal providing hierarchical access for supervisors and HR personnel based on secure role-based access control permissions.

The functional scope extends to employee centric tools such as a self-evaluation portal and goal management modules. These features allow staff to view their own productivity metrics in real time and track progress against departmental milestones. Additionally, the system incorporates attendance monitoring through digital login sessions and generates quantifiable reports on individual and team performance trends. From a data perspective, the study includes a secure database to store activity logs and historical productivity trends to support performance analysis within the MTN Rwanda corporate network.

To ensure feasibility and the protection of employee rights, certain elements are explicitly excluded from this study. The research does not cover physical surveillance through cameras or microphones, nor does it involve the content analysis of private communications, emails, or personal messages. The system is strictly limited to activity classification and does not perform automated disciplinary actions or payroll processing. Furthermore, complex integrations with third party HR suites and the monitoring of personal mobile devices are excluded from this phase. By maintaining these boundaries, the study remains focused on enhancing digital productivity through intelligent and ethical monitoring.

Methodology and Techniques Used in the Study

This study employed a qualitative research methodology combining direct observation and informal interviews as primary data collection methods to gather information on current productivity monitoring practices at MTN Rwanda and to inform the design of ProTrackAI. The methodology that was used during this final year project was agile.

Agile Methodology

Agile methodology is an iterative and incremental approach to software development that prioritizes flexibility, continuous feedback, and rapid delivery of functional components. Unlike traditional linear models, Agile allows the developer to break the project into manageable cycles known as sprints, each resulting in a working version of a specific feature. This method is particularly effective for complex systems like ProTrackAI, where requirements may evolve as stakeholders interact with initial prototypes. By focusing on collaboration and adaptive planning, Agile ensures that the final product is both technically robust and highly relevant to the end user's needs.

In the development of this project, Agile methodology was utilized to manage the design and implementation phases through a series of structured sprints. Each sprint focused on a specific core module, such as the activity tracking engine or the reporting dashboard, followed by a review session with stakeholders at MTN Rwanda. This iterative process allowed the researcher to incorporate feedback immediately, ensuring that the classification algorithms and user permissions were fine tuned to meet actual departmental standards. By using this approach, the system was built progressively, allowing for constant testing and validation of features like the self-evaluation portal before the final deployment phase was reached.

Observation

Observation is a systematic data collection method where the researcher examines specific behaviors and events within their actual setting. In the field of software engineering, this approach is vital for identifying the functional realities of a workplace without relying on secondary reports. By maintaining a non-intrusive presence, the investigator can gather objective

data regarding manual processes and system interactions, which serves as the primary foundation for identifying technical gaps in the current infrastructure. This technique ensures that the requirements gathered are based on actual user behavior rather than theoretical assumptions.

During this study, observation was employed at MTN Rwanda to monitor the digital workflows and the existing internet management protocols. The researcher spent time observing the limitations of the current portal, specifically how it functions as a basic firewall rather than a performance tracking tool. It was noted that the lack of automated classification leads to significant time being spent on non-work activities without any administrative awareness. This process allowed for a clear understanding of the operational bottlenecks and the specific user interface needs that the proposed ProTrackAI system intends to resolve.

Interviews

Interviews are a qualitative research method involving structured or semi-structured discussions with key stakeholders to obtain detailed insights into their experiences, perceptions, and operational challenges. This method provides practical data that complement findings from observation, allowing the researcher to explore the underlying motivations behind certain behaviors. Through these direct interactions, the investigator can capture the nuanced needs of the organization that are often missed by automated logs or passive observation, ensuring the final technical solution addresses the real world needs of the management team.

In this study, I conducted interviews with the IT Managers, Department Heads, and various staff members at MTN Rwanda. These discussions provided an in depth understanding of the current operational processes and highlighted the critical need for a more centralized, intelligent, and automated productivity system. The interviews focused specifically on operational bottlenecks, communication gaps, difficulties in performance planning, and the overarching expectations from a modern digital monitoring platform.

I, Jesse Shema Kyanga, a finalist in the Faculty of Information Technology, Department of Software Engineering at AUCA, conducted this research to design and develop the ProTrackAI system. This project specifically addresses inefficiencies in employee digital monitoring,

resource allocation, and real-time productivity analytics within the MTN Rwanda corporate environment.

Below are the key questions asked during the interviews, along with summarized responses:

Question 1: Does the current monitoring system actively block certain websites or restrict employee access?

Answer: "The capability to block websites exists within the current system, but it is not actively utilized. Employees generally have unrestricted access to social media and video platforms as long as the content is not deemed inappropriate."

Question 2: How is internet usage currently reviewed if there is no active blocking or analysis happening?

Answer: "Management occasionally monitors the logs, but there is no real-time analysis or automated system to distinguish between productive work and personal browsing. We essentially rely on the honor system for day to day operations."

Question 3: What operational challenges affect the accurate tracking of employee task completion?

Answer: "Currently, we rely on manual updates and end-of-week reports, which are often subjective and do not reflect the actual time spent on specific digital tasks or the contextual use of the internet."

Question 4: How would a real-time self-evaluation portal assist employees in managing their own professional development?

Answer: "Providing employees with direct access to their own analytics would foster a culture of accountability, allowing them to adjust their habits autonomously and prepare more effectively for performance reviews."

Expected Results

The implementation of the ProTrackAI system is expected to generate significant improvements in how MTN Rwanda supports, monitors, and empowers its digital workforce. The system is designed to modernize the existing oversight framework by enhancing accessibility, operational efficiency, and data driven transparency within the corporate environment. The following measurable improvements are anticipated upon the full integration of the software into the daily operational workflow

Core System Capabilities:

- Real-time tracking and AI-powered classification of employee digital activities across web and desktop applications
- Accurate distinction between productive work engagement and non-productive distractions
- Contextual understanding that differentiates work-related content from recreational use on the same platforms

Productivity Reporting and Analytics:

- Comprehensive productivity reports generated at individual, team, and departmental levels
- Daily, weekly, and monthly metrics on time utilization, application usage patterns, and productivity trends
- Data-driven insights to replace subjective assessments in performance evaluations
- Informed decision-making support for resource allocation, training needs, and employee development

Accountability and Transparency:

- Transparent goal-setting and progress-monitoring frameworks fostering organizational accountability
- Clear performance expectations with objective measurement of progress toward defined objectives
- Recognition systems for employees maintaining strong work discipline
- Identification of employees requiring additional support or training

Employee Empowerment:

- Self-evaluation portal providing workers with insights into their own productivity patterns
- Real-time feedback on digital work habits, time allocation, and focus patterns
- Personal development support reducing procrastination behaviors observed during the internship
- Transparency that benefits high-performing employees through objective recognition

Privacy and Security:

- Hierarchical access controls ensuring appropriate data visibility based on organizational roles
- Privacy-conscious design aggregating sensitive information for management reports
- Detailed tracking maintained for meaningful analysis while protecting employee privacy

Organizational Impact:

- Significant reduction in unproductive time wastage (15-22% productivity improvement based on similar systems)
- Elimination of the observed gap where employees could spend entire days on non-work activities without accountability
- Data-driven workforce management foundation supporting strategic HR decisions
- Long-term improvements in employee engagement and organizational effectiveness in Rwanda's telecommunications sector

Real-time Performance Visibility:

- Managers will have access to live dashboards providing immediate insights into departmental time allocation, allowing for proactive intervention before project deadlines are impacted.

Organization of Work

Chapter 1: This chapter establishes the foundation for the study by presenting the background of digital productivity challenges at MTN Rwanda. It outlines the problem statement, highlighting the limitations of current content filtering versus active productivity management. The chapter defines research motivations, objectives, and the study scope. Additionally, it details the methodology, covering the observation and interview techniques used during the internship alongside the expected results.

Chapter 2: This chapter provides a comprehensive evaluation of the existing productivity monitoring framework. It begins with an introduction to systems analysis principles and describes the current portal based environment and its operational deficiencies. The AS-IS process model is used to visualize existing workflows and monitoring practices. The chapter concludes by proposing technical solutions and outlining the functional and non-functional requirements for the new system.

Chapter 3: This chapter focuses on the technical architecture of ProTrackAI using Unified Modeling Language standards. It presents detailed UML diagrams, including use case, class, and sequence diagrams to illustrate system interactions and object oriented structures. Database design is covered through entity relationship diagrams and data dictionaries. The chapter also illustrates the overall system architecture, including the desktop agent, backend services, and the AI classification module.

Chapter 4: This chapter documents the practical development and validation of the ProTrackAI platform. It details the technology stack, including React, Node.js, and MongoDB. Visual evidence is provided through system screenshots, followed by a description of testing procedures and the hardware and software requirements for deployment.

Chapter 5: The final chapter synthesizes the research findings and evaluates the extent to which objectives were accomplished. It summarizes the achievements of ProTrackAI and provides recommendations for system deployment and user training. The chapter concludes with suggestions for future enhancements and broader system integrations.

CHAPTER 2

ANALYSIS OF CURRENT SYSTEM

Introduction

Systems analysis is a critical discipline in information technology that focuses on evaluating existing processes, identifying inefficiencies, and designing improved solutions (Kendall & Kendall, 2013). It involves the systematic examination of organizational operations, emphasizing information flow, resource utilization, and the achievement of business objectives. Through structured analysis techniques, researchers identify gaps between current practices and desired outcomes, providing the essential foundation for effective system design and implementation.

In the context of productivity management, systems analysis is vital for understanding how organizations monitor and optimize workforce performance. Traditional oversight often fails to leverage modern technological capabilities, resulting in missed opportunities for data driven decision making. By conducting a thorough analysis of existing systems, researchers can identify specific pain points and propose targeted solutions that address root causes rather than symptoms. This framework also establishes baseline performance metrics and justifies investment in new technologies by documenting how proposed solutions deliver measurable improvements.

For this final year project, systems analysis served as the foundation for evaluating the productivity monitoring approach at MTN Rwanda. To build an intelligent solution, this chapter examines the existing monitoring infrastructure through several key dimensions, including organizational context, strategic objectives, and a description of the current portal based system. Furthermore, the study utilizes AS-IS process diagrams and a multi-dimensional problem analysis to evaluate deficiencies in performance, data management, and operational efficiency. This comprehensive evaluation establishes the baseline necessary to design a system that resolves identified gaps while meeting organizational needs for productivity optimization and accountability.

Description of Current System Environment

Historical Background

MTN Rwanda is a leading telecommunications provider, serving millions of subscribers with mobile voice, data, and digital services. As a cornerstone of the national digital economy, the organization manages a diverse workforce across customer service, network operations, and administrative functions. The company's growth is closely aligned with Rwanda's rapid technological advancement and the increasing demand for robust digital infrastructure.

The nature of these operations requires employees to spend significant portions of their workday accessing internet resources for technical troubleshooting and system monitoring. This digital-intensive environment creates inherent challenges in distinguishing productive engagement from non-work activities. Despite its external technological sophistication, the organization's internal approach to productivity monitoring has remained relatively rudimentary, relying primarily on basic content filtering rather than intelligent activity analysis.

Mission

MTN Rwanda's mission centers around leading digital solutions for Africa's progress and on delivering exceptional telecommunications services that connect people, enable businesses, and contribute to national development goals. The organization strives to maintain operational excellence through efficient service delivery, customer satisfaction, and continuous technological innovation. Central to this mission is the effective utilization of human capital, as employee productivity directly impacts service quality and customer experience.

Vision

The organization envisions becoming Rwanda's most trusted and innovative telecommunications provider, they believe that everyone deserves the benefits of a modern connected liferecognized for service reliability, technological leadership, and positive socioeconomic impact. Achieving this vision requires not only external service excellence but also internal operational efficiency, including optimal workforce productivity management that ensures employees can focus on high-value activities while maintaining work-life balance and job satisfaction.

Description of Current System

The current employee internet monitoring system at MTN Rwanda operates through a portal based access control mechanism. While the interface suggests a high level of oversight, the practical application of the system remains limited. The following key points outline the operational reality of the current infrastructure:

- **Authentication and Access Control:** Employees use unique credentials to authenticate through a central portal featuring usage warnings and countdown timers, which create a visual impression of monitoring that is not fully reflected in practice.
- **Passive Logging vs. Active Blocking:** Although the system architecture has the technical capability to filter content, these features are not actively deployed. Observation confirmed that employees maintain unrestricted access to social media and entertainment platforms.
- **Lack of Administrative Analysis:** Overseeing the system is the task of a single staff member focusing on technical access issues rather than data analysis. Browsing patterns remain largely untapped for management or productivity purposes.
- **Absence of Classification:** The system fails to distinguish between professional research and personal browsing. Technical tutorials are logged identically to entertainment, preventing the generation of actionable productivity intelligence.
- **Self-Regulated Productivity:** There are no automated alerts for extended non-work activity or notifications for supervisors. Consequently, digital accountability is entirely self-regulated without external mechanisms to discourage distractions.
- **Disconnected Evaluation:** Performance assessments rely on traditional output metrics rather than digital activity. This represents a missed opportunity for data driven optimization, as disciplined habits receive no formal recognition.

Ultimately, a significant disconnect exists between the system's technical potential and its actual utilization. It remains a dormant logging tool that fails to provide the oversight required for a modern, high-performance digital environment. This current operational gap highlights the necessity for an automated classification engine that provides actionable insights.

Analysis of the Current System

The current productivity monitoring system at MTN Rwanda, while technically capable of collecting internet usage data, serves primarily as a passive logging tool rather than an active management resource. Systematic evaluation during the two month observation period revealed significant gaps between the system's architectural potential and its actual operational utility. The following points provide a detailed analysis of the existing infrastructure and its functional limitations:

- **Non Analytical Data Collection Pipeline:** The system is designed to log comprehensive internet usage data including timestamps, IP addresses, and destination URLs. However, it performs no internal processing to classify these activities or identify productivity patterns. This results in a vast repository of raw data that lacks the necessary analytical layer to generate actionable insights for organizational leadership.
- **Underutilized Infrastructure Capabilities:** While the existing network architecture includes built-in functionality for website blocking and content filtering, these features remain dormant. Technical oversight confirms that the capability to restrict non work related content exists within the software, yet it is not deployed in practice, representing an underutilization of the organization's technological investment.
- **Uniform Traffic Processing Model:** The system treats all digital engagement identically, making no distinction between high value professional activities and non productive browsing. For example, time spent on technical research or professional tutorials is recorded in the same format as entertainment consumption. Without a contextual classification engine, the logged data provides an inaccurate reflection of actual work effort.

This analytical breakdown confirms that while the foundational hardware for data collection is present, the existing system lacks the intelligent processing required to transform raw internet logs into a meaningful productivity management framework. This current operational gap highlights the necessity for an automated classification engine that provides actionable insights rather than raw data. By implementing such a solution, the organization can transform its passive logs into a proactive tool for enhancing overall workforce engagement and performance

Modeling of the current system

The existing operational workflow of MTN Rwanda's internet monitoring is visualized in Figure 1, which presents a formal AS-IS process model using a standard flowchart. This diagram illustrates the sequential progression from user authentication to data storage, highlighting the current system's linear and passive nature. A critical analysis of this model reveals the complete absence of decision nodes for automated traffic classification, real-time alerts, or productivity analysis. The process functions as a "dead-end" data pipeline where all internet traffic, regardless of its content, is treated identically and simply logged in a centralized database without subsequent review or actionable output for management.

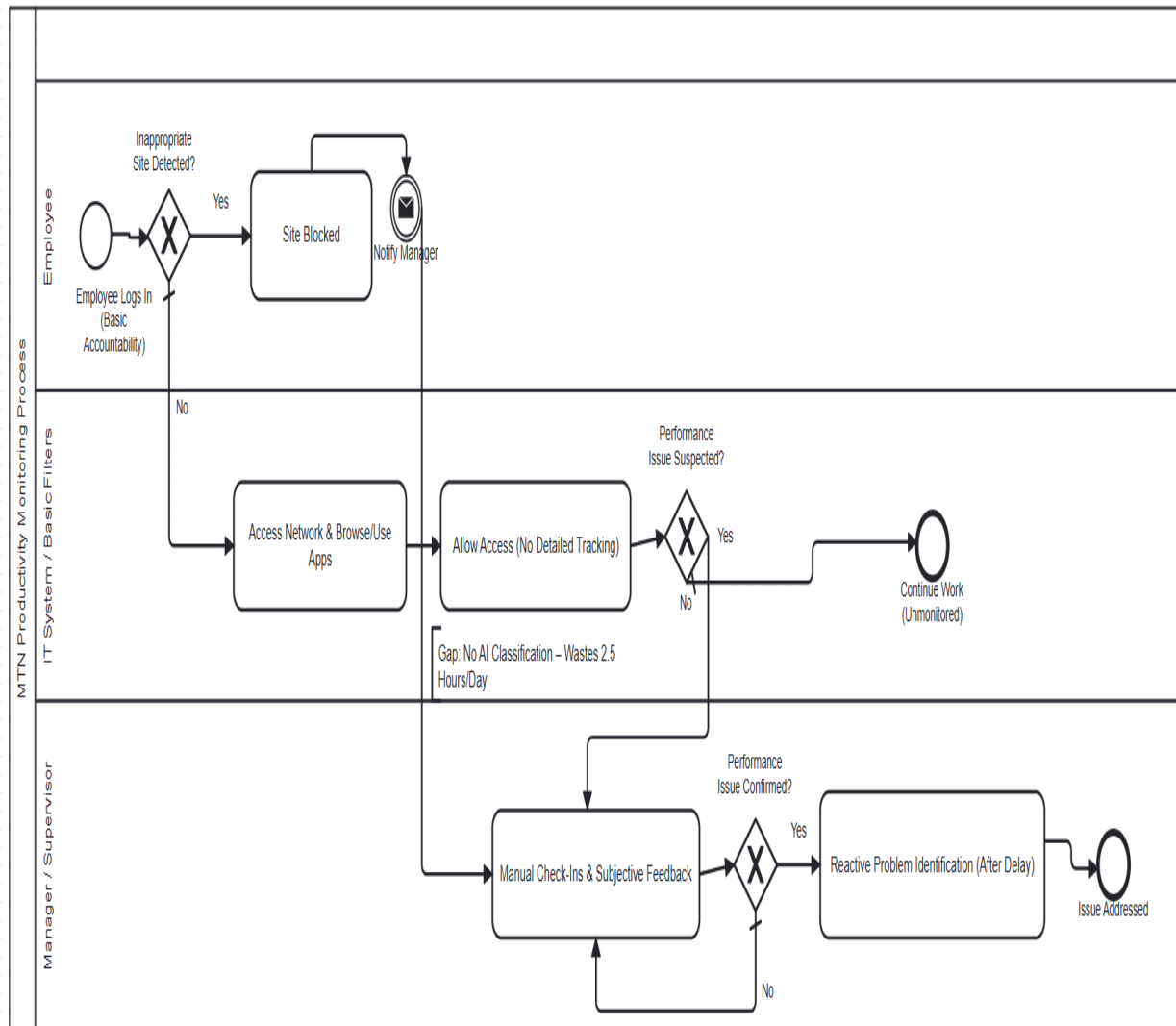


Figure 1: AS-IS process model

Problems of the Current System

The key problems of the current system at MTN Rwanda stem from its reliance on passive internet logging and content filtering, which lacks the intelligent capability to distinguish between productive and non-productive digital behavior. These limitations have significantly hindered management's ability to oversee employee performance effectively, provide objective feedback, and optimize workforce productivity. Without an automated and context-aware system, the organization remains vulnerable to undetected time-wastage and security risks, while managers lack the data-driven insights necessary for informed decision-making and personnel development.

Performance

Throughput:

- The existing system operates as a passive logger that cannot differentiate between work-related and leisure-based website usage, leading to undetected productivity gaps.
- The absence of real-time alerts and automated tracking further reduces the speed and effectiveness of management interventions.

Response time:

- The absence of real-time alerts and automated tracking reduces the speed and effectiveness of management interventions.
- Performance feedback is delayed because it relies on manual log reviews and subjective end-of-week reports.

Information

Input:

- Data collection is limited to basic URLs and timestamps, which lack the contextual classification or metadata needed to determine actual "time-on-task".
- The system relies on raw internet logs that do not capture the intent behind the digital activity.

Output:

- There is a complete absence of automated productivity reports or visual dashboards, forcing managers to rely on subjective observations
- Reports are often manually compiled and lack the analytical depth required to identify long-term productivity trends.

Storage:

- The system does not maintain detailed records of past internet activity in a way that allows for historical usage analysis.

Economics**Complexity:**

- Maintaining an extensive monitoring infrastructure that provides minimal productivity intelligence results in high operational costs with little return on investment.
- The reliance on manual oversight by a single staff member introduces significant labor expenses and creates a bottleneck that limits organizational scalability.

Control**Security:**

- The system provides no real-time monitoring of security threats or unauthorized digital activities, creating a management blind spot.
- While website-blocking capabilities exist, they are not actively deployed, leading to uncontrolled access to non-work-related platforms.

Efficiency

- Manual log reviews create a labor-intensive workflow that prevents timely feedback and hinders organizational agility.
- Employees may spend significant work hours on social media or video platforms without any automated mechanism to flag or discourage these distractions.

Service

- Managers lack objective performance data, making it difficult to accurately identify training needs or recognize high performers.

Proposed Solutions

Intelligent Activity Classification: Unlike the current system that logs all internet activity identically, ProTrackAI employs artificial intelligence algorithms to classify digital activities based on productivity value. The system distinguishes between work-related content consumption and personal entertainment, recognizes when commonly recreational platforms like YouTube are being used for professional development, and generates productivity scores that reflect genuine work engagement rather than mere online presence.

Automated Productivity Analytics: ProTrackAI transforms raw activity logs into comprehensive productivity reports at individual, team, and departmental levels. Management dashboards provide visualizations of work patterns, time allocation trends, and performance metrics. Automated report generation occurs daily, weekly, and monthly, eliminating the manual effort currently required for productivity assessment while delivering far more accurate and detailed insights.

Transparent Accountability Framework: The system establishes clear accountability mechanisms that benefit both employees and management. Workers gain access to self-evaluation portals showing their own productivity patterns, enabling personal development and time management improvement. Managers receive objective performance data to support fair evaluations, recognize high performers, and identify employees requiring additional support or training.

Goal Management & Progress Tracking: ProTrackAI implements structured goal-setting capabilities where supervisors can define performance objectives and track progress through quantifiable metrics. This replaces subjective assessment with data-driven evaluation, creating transparency around expectations and achievement that fosters continuous improvement culture.

Privacy-Conscious Design: While addressing the accountability gaps in the current system, ProTrackAI maintains employee privacy through thoughtful design. The system tracks application usage and website domains rather than content, aggregates detailed data for management reporting, and provides role-based access controls ensuring supervisors see only information relevant to their teams. This approach balances organizational oversight needs with employee dignity and trust.

System Requirements

Functional Requirements

The ProTrack AI platform must meet the following functional requirements to ensure effective productivity monitoring, objective reporting, and seamless management oversight at MTN Rwanda:

- **REQ 1:** The system shall track active applications and URLs in real-time during work hours and utilize AI algorithms to categorize activities as productive, non-productive, or neutral.
- **REQ 2:** The system shall generate objective productivity scores based on time allocation and provide employees with self-evaluation tools to view their own daily, weekly, and monthly activity summaries.
- **REQ 3:** The system shall provide customized real-time dashboards for Supervisors and HR Personnel to monitor team metrics and identify organizational productivity trends.
- **REQ 4:** Supervisors shall be able to set performance goals, monitor achievement progress, and provide direct feedback to employees through the system's communication interface.
- **REQ 5:** The system shall generate automated productivity reports and support data export in standard formats (CSV/PDF) for strategic workforce planning and analysis.
- **REQ 6:** System administrators shall manage user accounts and configure Role-Based Access Control (RBAC) to ensure that sensitive productivity data is only accessible to authorized personnel.
- **REQ 7:** Administrators shall be able to adjust AI classification parameters and system-wide policies to align with the organization's evolving definitions of productive activity.
- **REQ 8:** The system shall maintain comprehensive audit logs of all system access and send automated notifications for significant events, such as goal completion or extended inactivity.

Non-Functional Requirements

Performance and Scalability

- **REQ 9:** The system shall support at least 500 concurrent users without performance degradation and be architected for horizontal scaling to accommodate organizational growth.
- **REQ 10:** Dashboards shall load within 3 seconds, and AI data processing must be completed within 5 minutes of collection to ensure near-real-time visibility.

Reliability

- **REQ 11:** The system shall maintain 99.5% uptime during business hours and implement automated daily backups and fault-tolerant architecture to prevent data loss.

Security and Privacy

- **REQ 12:** The system shall utilize TLS encryption for data transmission, industry-standard password hashing, and data anonymization techniques to protect employee privacy and data integrity.

Usability

- **REQ 13:** The user interface shall be intuitive, requiring no more than three clicks for primary functions, and must be fully responsive across all standard desktop resolutions.

Compatibility

- **REQ 14:** The monitoring agent shall support Windows and Linux operating systems, while the web interface must function correctly across all modern browsers including Chrome, Firefox, Edge, and Safari.

Compliance

- **REQ 15:** The system shall comply with applicable data privacy regulations by providing transparent monitoring disclosures and supporting individual data rights.

CHAPTER 3

REQUIREMENTS ANALYSIS AND DESIGN OF THE NEW SYSTEM

Introduction

Requirements analysis and design represent the critical bridge between identifying organizational problems and implementing technical solutions. These phases transform abstract needs into concrete system specifications, establishing the architectural blueprint that guides development efforts. Through systematic analysis of stakeholder requirements and careful design of system components, software engineers ensure that implemented solutions effectively address real-world challenges while remaining technically feasible and operationally sustainable.

In this chapter, the focus shifts from understanding MTN Rwanda's existing productivity monitoring limitations to designing ProTrackAI as a comprehensive solution. Requirements analysis captures what the system must do the specific capabilities needed to support employees, supervisors, HR personnel, and administrators in their respective roles. System design addresses how these requirements will be implemented through technical architecture, component interactions, data structures, and user interfaces.

The chapter employs Unified Modeling Language (UML) as the primary tool for visualizing system design. UML provides standardized notation for representing complex software systems, enabling clear communication between technical and non-technical stakeholders. Through various diagram types including use case, class, sequence, activity, and database diagrams, the design specifications for ProTrackAI are documented in forms that support both implementation by developers and validation by domain experts. Scalability was held in consideration when

The design process follows industry best practices including iterative refinement, stakeholder validation, and consideration of non-functional requirements alongside functional capabilities. By documenting design decisions comprehensively, this chapter creates a reference point for development activities, testing validation, and future system maintenance or enhancement efforts.

Unified Modelling Language (UML)

Unified Modeling Language (UML) is a standardized visual modeling notation used extensively in software engineering to represent system designs, architectures, and behaviors. Developed in the mid-1990s and maintained by the Object Management Group (OMG), UML provides a common language for software developers, system architects, business analysts, and stakeholders to communicate about complex systems without ambiguity.

Purpose and Benefits

UML serves multiple critical functions in software development:

Visualization: Complex system architectures and component interactions become comprehensible through visual diagrams. Rather than parsing thousands of lines of descriptive text, stakeholders can quickly understand system structure and behavior through intuitive graphical representations.

Specification: UML diagrams serve as precise specifications for system functionality. Developers can reference these models during implementation to ensure the code aligns with design intent. The visual notation reduces misinterpretation compared to natural language descriptions.

Documentation: UML creates permanent documentation of system design that remains valuable throughout the software lifecycle. When systems require maintenance or enhancement years after initial development, UML diagrams provide crucial insight into original design decisions and system organization.

Communication: UML bridges communication gaps between technical and non-technical stakeholders. Business users can validate use case diagrams representing system functionality even without programming knowledge, while developers gain precise technical specifications from class and sequence diagrams. By providing a standardized visual framework, UML ensures that architectural decisions and system workflows remain transparent across diverse project teams. This alignment ultimately reduces the risk of costly rework by identifying logic gaps or requirement misunderstandings early in the development lifecycle.

UML Diagram Categories

UML encompasses numerous diagram types, broadly categorized as structural diagrams and behavioral diagrams:

Structural Diagrams represent the static architecture of systems:

- **Class Diagrams** show object-oriented classes, their attributes, methods, and relationships
- **Component Diagrams** illustrate high-level software components and their dependencies
- **Deployment Diagrams** depict how software is distributed across hardware infrastructure
- **Package Diagrams** organize related classes into logical groupings

Behavioral Diagrams represent dynamic system behavior:

- **Use Case Diagrams** capture functional requirements from user perspectives
- **Sequence Diagrams** show how objects interact over time through message exchanges
- **Activity Diagrams** represent workflows and business processes
- **State Machine Diagrams** model how objects transition between different states

For ProTrackAI's design documentation, this chapter focuses on five essential diagram types that comprehensively capture the system's requirements and architecture: use case diagrams for functional requirements, class diagrams for object-oriented structure, sequence diagrams for component interactions, activity diagrams for process workflows, and database diagrams for data persistence design.

UML in Agile Development

While UML originated in traditional waterfall development methodologies, it remains valuable in modern agile contexts. Rather than creating exhaustive upfront documentation, agile teams use UML diagrams selectively to clarify complex aspects of system design, communicate architectural decisions, and onboard new team members. ProTrackAI's UML documentation strikes this balance providing sufficient detail to guide implementation while avoiding excessive documentation overhead.

Design of the New System

The design of ProTrackAI employs systematic modeling techniques to transform identified requirements into technical specifications that guide implementation. This section presents comprehensive UML diagrams capturing the system's functional capabilities, architectural structure, component interactions, process workflows, and data persistence mechanisms.

Use Case Diagram

Use-Case Diagrams represent the functional requirements of ProTrackAI by mapping how users and system components interact to accomplish specific tasks. During development, these visual models were essential for translating the productivity gaps observed at MTN Rwanda into distinct features for employees, supervisors, and HR administrators. By providing a clear blueprint of the system architecture from a user perspective, these diagrams ensured that the implementation remained focused on solving real-world tracking needs while verifying that all original project objectives were fulfilled.

Use Case Diagram Components

Actor

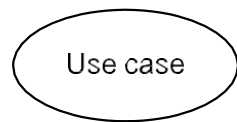
An actor in the diagram represents anyone or anything that interacts with ProTrackAI from outside the system boundaries. For this project, actors include three types of human users: regular employees whose productivity is tracked, supervisors who manage teams and review performance, and HR administrators who oversee the entire organization. There are also two automated actors: the desktop agent software running on employee computers that captures activity data, and the AI classification engine that analyzes activities to categorize them. Actors are drawn as stick figures in UML notation, with their specific role label written underneath.



Use Case

Each use case represents a complete task that an actor can perform using ProTrackAI. For example, logging into the system, viewing productivity reports, or assigning performance goals are all distinct use cases. A use case defines what the system must do to enable that task, including what happens step by step when someone performs it. In UML diagrams, use cases

appear as labeled ovals inside the system boundary box. They describe capabilities from the user's point of view rather than technical implementation details, making them valuable for understanding what the system accomplishes regardless of how the code works internally.



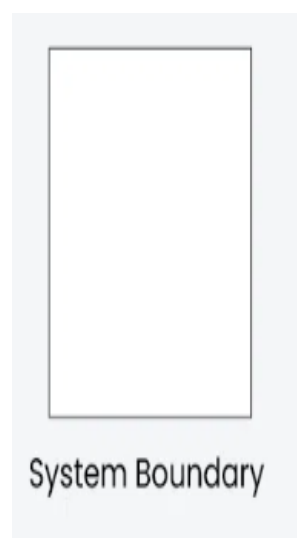
Relationship

Relationships connect actors to use cases via solid lines, indicating participation in a specific functionality. In ProTrackAI, these links differentiate access levels, such as allowing employees to view personal reports while restricting team management features to supervisors. Additionally, dashed arrows represent dependencies where one use case 'triggers' or 'feeds' data to another, ensuring a seamless automated workflow.



System Boundary

A rectangular boundary box separates ProTrackAI's internal functionality from its external actors. This clarifies the project scope, distinguishing implemented features from the users and components that interact with them. For example, while the desktop agent and AI classifier initiate processes, they remain external to the system's core boundary.



Use Case Diagram

The Use-Case Diagram for ProTrack AI illustrates the functional interactions between the system and its primary stakeholders: Employees, Supervisors, and HR Personnel. It maps out how automated background processes provide the data foundation for user-driven features such as productivity monitoring, goal setting, and administrative oversight.

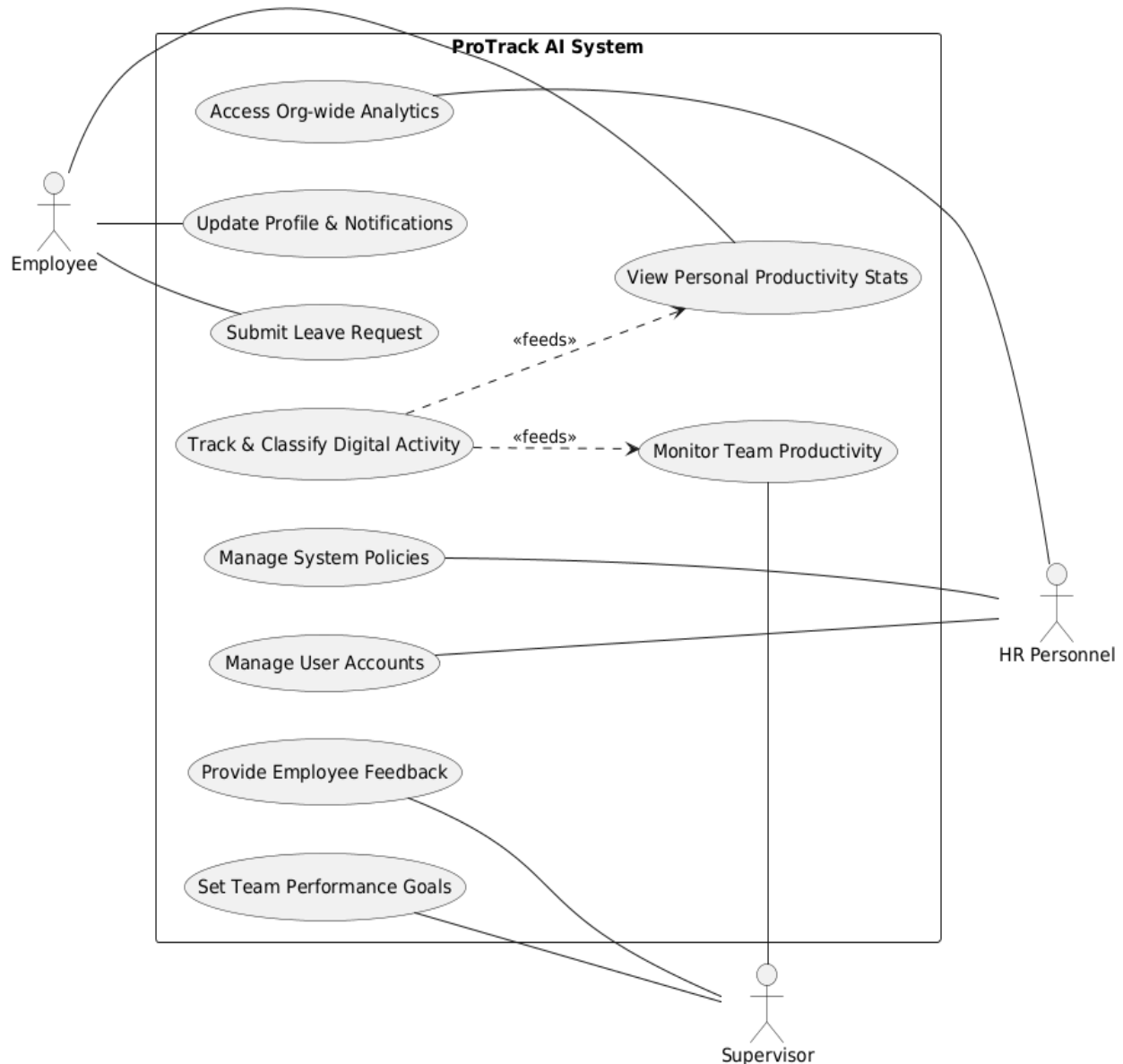


Figure 2: Use Case Diagram

Use Case Description

Each use case in ProTrackAI is documented with standardized details that completely describe how that functionality works. These descriptions follow a consistent format with seven specific components that together explain the complete behavior:

- **Name:** The specific title given to the use case, chosen to clearly indicate what action or capability it represents.
- **Description:** A summary explaining what this use case accomplishes and why it matters for productivity management at MTN Rwanda.
- **Actor(s):** The specific roles that can initiate or participate in this use case, such as employees, supervisors, HR administrators, or automated system components.
- **Pre-condition:** The situation that must already exist before this use case can start, such as the user being logged in or certain data already being stored in the database.
- **Post-condition:** The new situation that exists after the use case completes successfully, describing what changed in the system or what information is now available.
- **Normal Flow:** The typical sequence of steps that occurs when everything works as expected, numbered in order from start to finish.
- **Alternative Flow:** What happens if something goes wrong or if the user makes a different choice, describing exceptions and error handling.

The following tables document each major use case in ProTrackAI using this standardized format:

Use Case	Login
Actor(s):	Employee, Supervisor, HR, Administrator
Description:	Allows authenticated users to access the ProTrackAI system based on their assigned organizational roles. The system validates credentials and

	grants access to role-specific dashboards and features.
Pre-condition:	User has valid credentials registered in the system database
Post-condition:	User is authenticated and redirected to their role-based dashboard with appropriate access permissions
Normal Flow:	1. User navigates to the login page and enters email or employee ID along with password 2. System validates credentials against the database using bcrypt password comparison 3. System generates a JSON Web Token containing user role, team, and identification information 4. System stores authentication token in browser local storage 5. User is redirected to role-specific dashboard (Employee, Supervisor, or HR view)
Alternative Flow:	If credentials are invalid, system displays error message and prompts user to retry

Table 1: Login Use Case

Use Case:	Track Digital Activity
Actor(s):	Desktop Agent (System)
Description:	Automated process where the desktop agent continuously monitors and captures employee digital activity including active applications, window titles, and browser URLs during work hours. This is a core system function that operates transparently in the background.
Pre-condition:	Desktop agent is installed and running on employee workstation; Employee is logged into the agent; Employee has clocked in for the day

Post-condition:	Activity metadata is captured and transmitted to the backend server for processing and storage
Normal Flow:	1. Desktop agent runs in system tray and monitors active window every thirty seconds 2. Agent captures application name using operating system APIs 3. Agent extracts window or tab title from active window 4. For browser windows, agent captures active URL from address bar 5. Agent packages data with timestamp and employee identifier 6. Agent transmits activity payload to backend via POST request to /api/activities/raw endpoint 7. Agent schedules next capture cycle after thirty seconds
Alternative Flow:	If backend server is unreachable, agent retries connection; After twenty consecutive failures, agent gracefully shuts down to conserve system resources

Table 2: Tracking Use Case

Use Case:	Classify Activity
Actor(s):	AI Classifier (System)
Description:	Automated intelligent classification of captured digital activities as productive, non-productive, or neutral using a multi-layer decision cascade powered by TensorFlow.js neural networks and rule-based logic. This process provides the intelligence layer of the productivity tracking system.
Pre-condition:	Raw activity data has been received from desktop agent and stored in temporary processing queue
Post-condition:	Activity is classified with a confidence score and stored in MongoDB activity collection

Normal Flow:	1. AI module receives activity metadata containing application name, window title, and URL 2. System preprocesses text by cleaning application names and extracting domain from URLs 3. AI applies role-based exceptions for marketing and HR roles using social media tools professionally 4. System checks against known entertainment applications database 5. AI examines custom weighted rules defined by supervisors 6. System performs domain-level classification for recognized websites 7. AI evaluates strong work-related keywords and productivity indicators 8. If no high-confidence match found, TensorFlow.js neural network performs contextual classification 9. System calculates confidence score and assigns final classification 10. Classified activity with confidence score is stored in database
Alternative Flow:	If confidence score is below fifty percent, activity is flagged for manual review in supervisor queue

Table 3: Classification Use Case

Use Case:	View Productivity Report
Actor(s):	Employee, Supervisor
Description:	Allows users to view comprehensive productivity analytics showing time allocation, efficiency metrics, and activity breakdowns over selectable time periods. Employees view personal data while supervisors access team aggregations.

Pre-condition:	User is authenticated and has activity data recorded in the system
Post-condition:	Productivity report is displayed with charts, metrics, and activity summaries
Normal Flow:	1. User navigates to reports section from dashboard 2. User selects time period (today, week, or month) 3. System queries activity database filtering by user identifier and date range 4. Backend aggregates data calculating total time, productive time, neutral time, and non-productive time 5. System computes productivity percentage as ratio of productive to total time 6. Frontend displays metrics using doughnut charts and activity timelines 7. Recent activities are shown with classification labels and confidence indicators 8. Top productive applications are highlighted with usage statistics
Alternative Flow:	Supervisors viewing team reports see domain-level data instead of specific URLs to maintain employee privacy

Table 4: Productivity Report Use Case

Use Case:	Set Goals
Actor(s):	Supervisor
Description:	Enables supervisors to define performance goals for team members including target metrics, descriptions, and due dates. Goals are used to align individual performance with organizational objectives.
Pre-condition:	Supervisor is authenticated and has team members assigned in their

	organizational hierarchy
Post-condition:	Goal is created in database and assigned employee receives notification
Normal Flow:	1. Supervisor navigates to goal management interface 2. Supervisor clicks create new goal button 3. System displays goal creation form 4. Supervisor enters goal title, detailed description, and target outcome 5. Supervisor selects employee from team dropdown list 6. Supervisor sets due date using date picker 7. System validates all required fields are completed 8. Supervisor submits goal for creation 9. Backend stores goal in MongoDB with assigned employee identifier and creator identifier 10. System sends real-time notification to employee via Socket.IO 11. Employee sees new goal in their dashboard pending tasks section
Alternative Flow:	If employee identifier is invalid, system displays error message

Table 5: Goal setting use case

Use Case:	Submit Self-Evaluation
Actor(s):	Employee
Description:	Provides employees with a portal to conduct self-assessments, reflecting on their performance, accomplishments, and areas for improvement. This promotes self-awareness and supports the performance review process.
Pre-condition:	Employee is authenticated and has been prompted or scheduled for evaluation period

Post-condition:	Self-evaluation is submitted and marked as pending HR review in the system
Normal Flow:	1. Employee receives notification prompting self-evaluation submission 2. Employee navigates to evaluation portal from dashboard 3. System displays evaluation form with assessment questions 4. Employee writes self-assessment describing accomplishments and challenges 5. Employee identifies areas for improvement and professional development needs 6. Employee assigns self-rating on performance scale 7. System validates that all required fields contain meaningful responses 8. Employee submits completed evaluation 9. Backend stores evaluation with pending status in evaluations collection 10. HR personnel receive notification of new evaluation requiring review
Alternative Flow:	Employee can save draft evaluation and return to complete it later

Table 6: Self Evaluation Use Case

Use Case:	Generate Team Report
Actor(s):	Supervisor
Description:	Allows supervisors to generate comprehensive PDF reports summarizing team performance metrics including individual member productivity, task completion rates, and attendance statistics for management review and documentation.
Pre-condition:	Supervisor is authenticated and has team members with recorded activity data

Post-condition:	PDF report is generated and downloaded to supervisor's device
Normal Flow:	1. Supervisor accesses team dashboard and clicks export report button 2. System prompts supervisor to select reporting period (week, month, or custom range) 3. Backend queries all team member activity data within selected timeframe 4. System aggregates productivity percentages, total hours logged, and task completion rates per employee 5. Backend calculates team averages and identifies top performers 6. PDFKit library generates formatted document with MTN Rwanda branding 7. Report includes performance table with color-coded efficiency indicators 8. System computes team aggregates showing overall productivity and cumulative hours 9. PDF is streamed to client browser for download 10. Supervisor's device saves file as team performance report with timestamp
Alternative Flow:	If no team data exists for selected period, system notifies supervisor that insufficient data is available

Table 7: Generate Team Report Use Case

Use Case:	Manage Users
Actor(s):	Administrator (HR)
Description:	Provides HR administrators with capabilities to update employee information and manage the organizational structure. This primarily involves reassigning employees to different teams or departments and updating their position details to ensure accurate reporting.
Pre-condition:	Administrator has HR role privileges in the system and user accounts already

	exist
Post-condition:	User account details (team, department, position) are updated in the database
Normal Flow:	1. HR administrator navigates to the organization management interface 2. Administrator searches for a specific employee in the user directory 3. Administrator selects the user to view their current profile 4. Administrator updates the employee's team assignment, department, or position title 5. System validates the changes 6. Backend processing updates the user record in the MongoDB users collection 7. Any associated active tasks or reports are automatically linked to the new team context
Alternative Flow:	If an Administrator attempts to update a user ID to one that already exists, the system rejects the change to prevent data conflicts

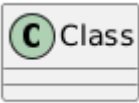
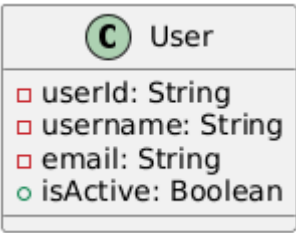
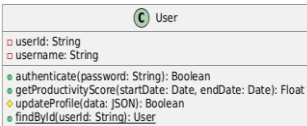
Table 8: HR managing users

Class Diagram

Class diagrams serve as a fundamental structural blueprint for ProTrackAI, providing a static snapshot of the system architecture and its object-oriented design. By mapping out essential classes such as User and ActivityRecord, these diagrams define the specific data fields and operations required to manage productivity metrics and account information. Explicitly defining relationships such as how specific user roles inherit from a base class or how activity logs associate with individual accounts ensured a coherent database schema and streamlined the development process. Ultimately, these visual models acted as a vital communication tool during design reviews, allowing for the early identification of logic gaps and providing a clear map of how data flows through the AI classification engine.

Class Diagram Components

The following table explains the key elements used in class diagrams and their meanings within the ProTrackAI system:

Component	Visual Representation	Description
Class		Represents a blueprint for objects in the system. The top section contains the class name, the middle section lists attributes (data fields), and the bottom section shows methods (functions). For example, the User class defines how all user accounts are structured.
Attributes		Define the data that each object of this class will store. The + symbol indicates public attributes accessible from outside the class, while - indicates private attributes only accessible within the class. For instance, +email: String means every user has a publicly accessible email field.
Methods		Specify the operations or behaviors that objects of this class can perform. Methods are functions that operate on the class data. For example, +authenticate() is a method that verifies user login credentials.

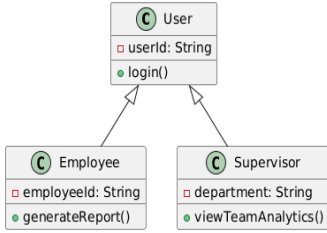
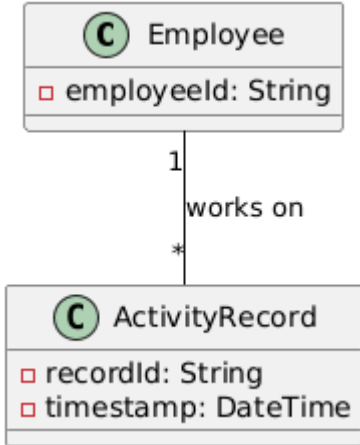
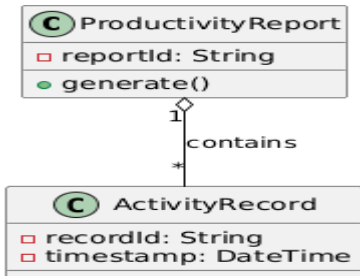
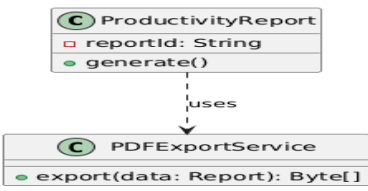
Inheritance	 <pre> classDiagram class User { +String userId +login() } class Employee { +String employeeId +generateReport() } class Supervisor { +String department +viewTeamAnalytics() } User < -- Employee User < -- Supervisor </pre>	Shows an "is-a" relationship where a child class inherits all attributes and methods from a parent class. The arrow points from child to parent. In ProTrackAI, Employee inherits from User, meaning employees have all User properties plus any additional employee-specific ones.
Association	 <pre> classDiagram class Employee { +String employeeId } class ActivityRecord { +String recordId +DateTime timestamp } Employee "1" -- "*" ActivityRecord : works on </pre>	Indicates a relationship where objects of one class are connected to objects of another. Numbers like 1 and 0..* show how many instances can be related. For example, User "1" -- "0..*" ActivityRecord means one user can have zero or many activity records.
Aggregation	 <pre> classDiagram class ProductivityReport { +String reportId +generate() } class ActivityRecord { +String recordId +DateTime timestamp } ProductivityReport "1" o-- "*" ActivityRecord : contains </pre>	Represents a "has-a" relationship where one class contains or is composed of others, but the parts can exist independently. The diamond appears on the container class. ProductivityReport aggregates ActivityRecords because reports are built from activities.
Dependency	 <pre> classDiagram class ProductivityReport { +String reportId +generate() } class PDFExportService { +export(data: Report): Byte[] } ProductivityReport ..> PDFExportService : uses </pre>	Indicates that one class uses or depends on another temporarily, typically by calling its methods or using it as a parameter.

Table 9: Class diagram tools Description

Sequence Diagram

Sequence diagrams illustrate ProTrackAI's dynamic behavior by capturing the specific chronological order of messages exchanged between components. They were critical for verifying complex workflows, such as the AI classification process, by mapping data hand-offs between the desktop agent and backend server. This temporal view helped identify potential bottlenecks and refine communication protocols, ensuring the system handles processing delays without impacting the user experience.

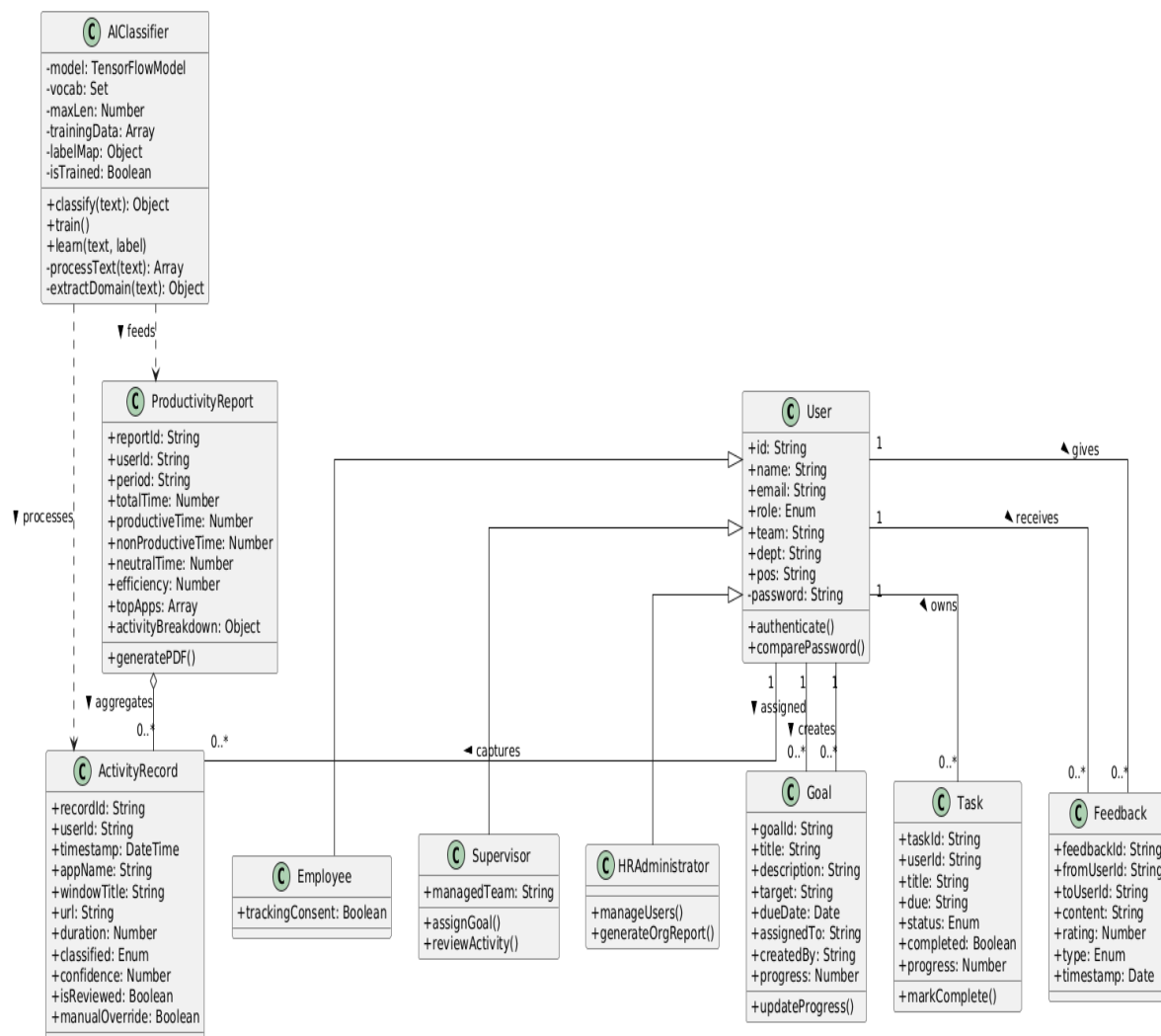


Figure 3: Class Diagram

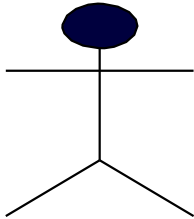
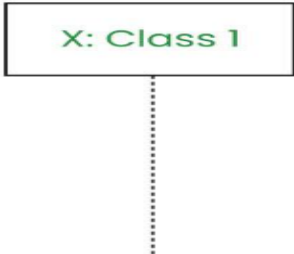
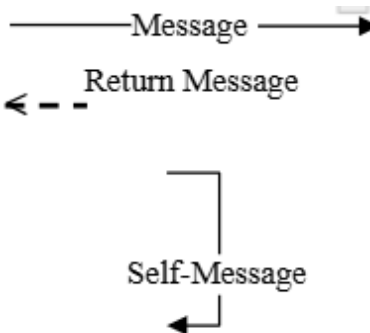
Component	Visual Representation	Description
Actor		Represents a human user (like an Employee) who initiates a sequence of events. Actors are placed at the top of the diagram.
Lifeline		Extends downward from each actor or participant, representing the passage of time. Interactions read from top to bottom.
Activation Bar		Indicates the period when a participant is actively processing a task or waiting for a specific return message. It shows the duration of an operation.
Message		A request where the sender waits for a response before continuing. For example, the frontend calling an API endpoint and waiting for data.

Table 10: Sequence diagram definition

Activity classification sequence diagram

This sequence diagram illustrates the automated workflow of ProTrack AI, detailing the interactions between the Monitoring Agent, the Backend Server, and the AI Classification Engine. The process begins with the Monitoring Agent capturing real-time digital activities and transmitting this raw data to the Backend Server. The server then invokes the AI Classification Engine, which analyzes the metadata to categorize the activity as productive, non-productive, or neutral. Finally, the classified data is returned and stored in the database, enabling the system to generate the accurate performance metrics and reports subsequently accessed by users.

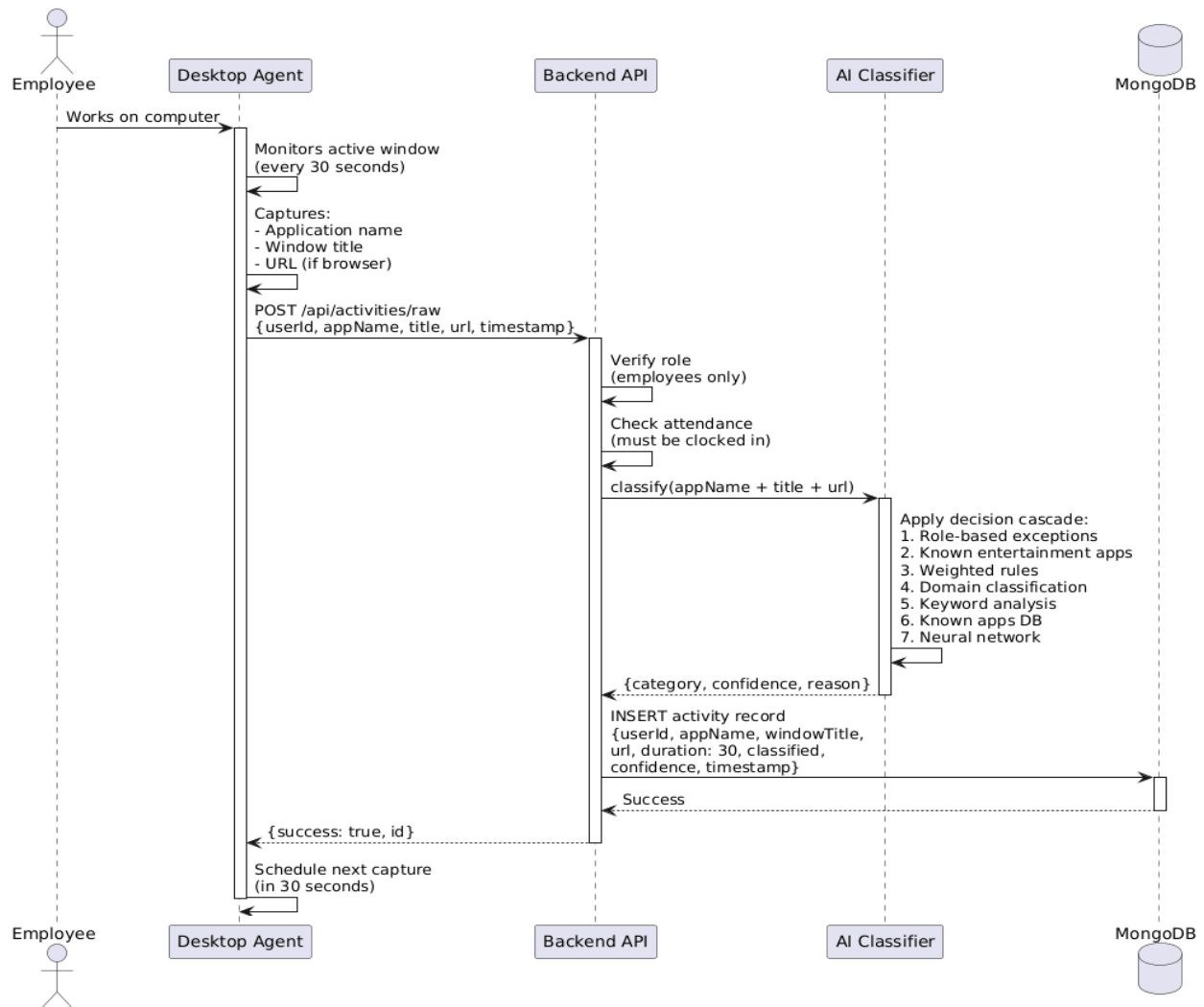


Figure 4: Activity Classification Sequence Diagram

Report generation sequence diagram

This sequence diagram outlines the process of report generation within the ProTrack AI system, triggered by an authorized user such as a Supervisor or HR Personnel. The workflow begins when the user requests a specific productivity report through the Web Interface, which sends a query to the Backend Server. The server then retrieves the relevant classified activity logs from the Database and processes this data to calculate aggregate metrics and performance scores. Once the analytical processing is complete, the Backend Server formats the information into a visual report or downloadable document, which is then served back to the Web Interface for the user to view or export.

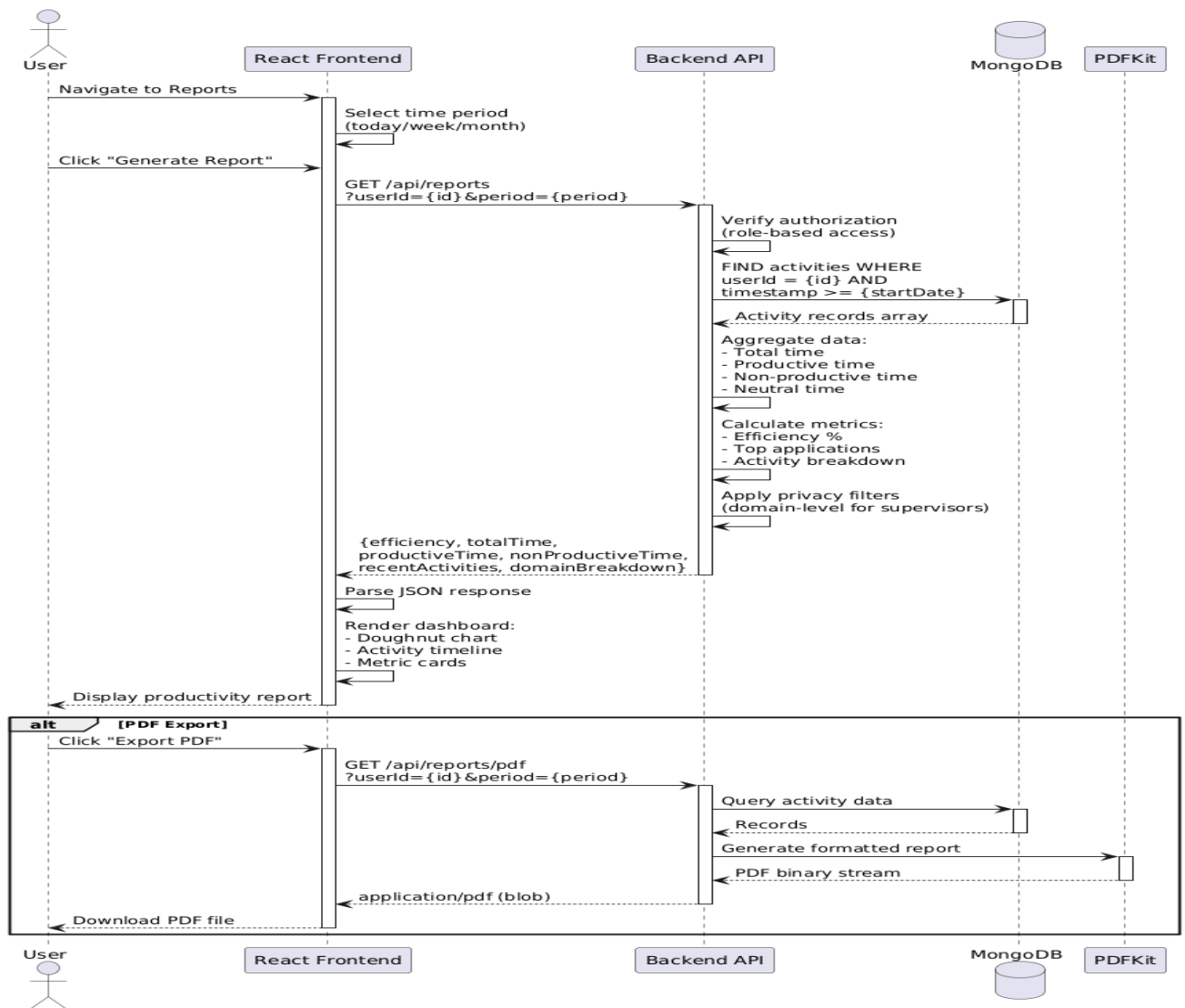


Figure 5: Report Generation Sequence Diagram

Database Diagram

The database diagram represents the foundational data structure of the ProTrackAI system, illustrating how information is organized, stored, and related. For this project, I moved away from traditional relational models to a document-oriented approach using MongoDB. This diagram visualizes the schema design, showing the various collections (equivalent to tables) and how they connect to one another to support the application's functionality.

Designing the database schema was a critical first step because it dictated how efficiently the system could handle high-volume activity logs and complex reporting queries. The diagram served as a blueprint for implementing the Mongoose models in the backend, ensuring that every data entity from simple user profiles to complex AI classification records had a defined structure while maintaining the flexibility to evolve. It also helped identify where indexes were needed to optimize performance for the real-time dashboards.

Database Structure Elements

At its core, the ProTrackAI database is built upon ten distinct collections, which serve as containers for related data records similar to tables in a relational database. Each record is stored as a document a flexible, JSON-like structure that holds specific fields such as names, dates, and status indicators. Every document is automatically assigned a unique `_id` acting as its primary key, ensuring precise identification.

To maintain data integrity across the system, references are used to link documents together. For example, an activity log contains a `userId` field that points to the specific employee document it belongs to, functioning effectively as a foreign key. The schema also defines specific data types for each field including strings for text, numbers for calculations, and enumerations for fixed choices like 'productive' or 'neutral' to ensure consistent data quality throughout the application.

This document-oriented approach provides the scalability needed to handle the high volume of real-time activity logs generated by the desktop agent without sacrificing query performance.

Database Diagram

Unlike traditional relational databases (SQL), MongoDB is a NoSQL document database and does not use physical foreign keys or enforced constraints. Consequently, the diagram above depicts independent collections rather than interconnected tables. The relationships between data entities (such as a User owning an Activity) are valid and critical, but they are "logical" relationships managed by the application layer rather than "physical" connections enforced by the database engine. This decoupled design was chosen specifically to maximize write performance for high-volume activity logging.

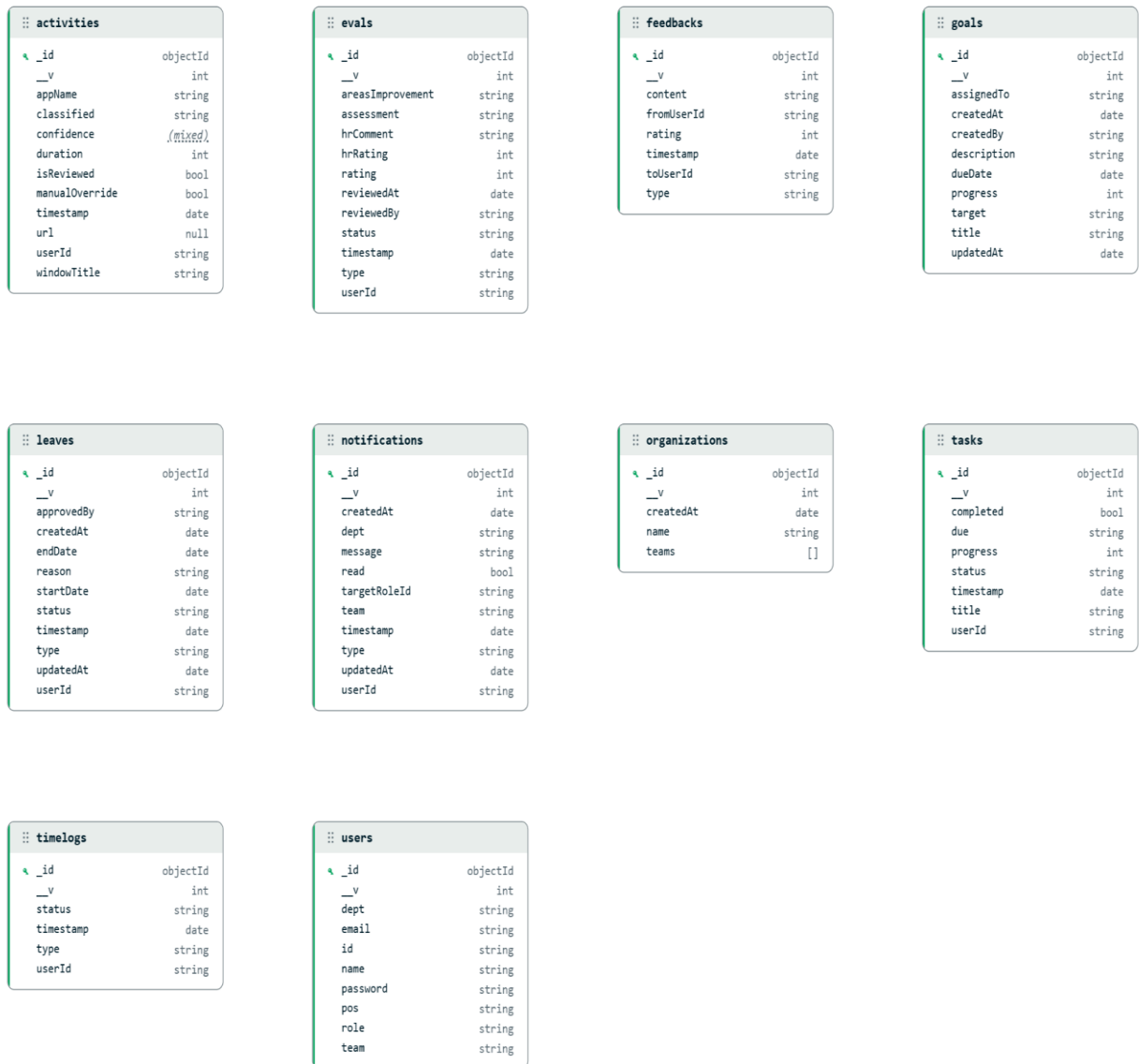


Figure 6: Database Diagram

System Architecture Design

System architecture design defines the overall structure and components of the ProTrackAI system to ensure it meets the functional and non-functional requirements of enhancing workplace productivity and monitoring. This design focuses on ensuring seamless connectivity between employee workstations and the central server, supporting real-time data processing for the AI classifier, facilitating secure communication between supervisors and their teams, and ensuring the privacy-conscious management of activity logs.

This section also illustrates the flow of how activity tracking, productivity analysis, and report generation work, including the backend processing of metadata, security measures for sensitive employee records, and the mechanisms for transparency in performance evaluation. Robust security frameworks are embedded to ensure data privacy, prevent unauthorized access to personal activity data, and maintain the integrity of performance metrics. Transparent and efficient communication between the desktop agents, the web dashboard, and the database is prioritized to ensure a smooth and responsive user experience across the organization.

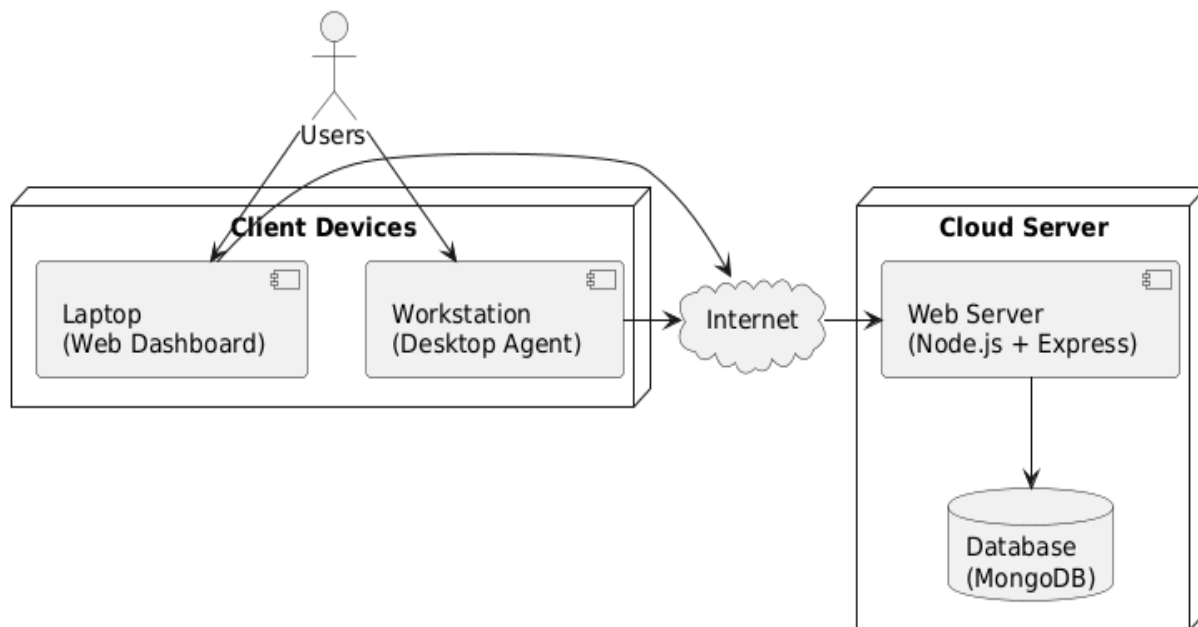


Figure 7: System Architecture Design

CHAPTER 4

IMPLEMENTATION AND TESTING OF THE NEW SYSTEM

Introduction

The implementation phase marked the transformation of the ProTrackAI architectural designs into a fully functional productivity management platform. This process followed an iterative development cycle spanning approximately twelve weeks, allowing for the progressive integration of the React frontend, Node.js backend, and the TensorFlow.js classification engine. The primary focus was on ensuring that the theoretical models for activity tracking and real-time analysis were successfully realized in efficient, production-ready code. This chapter details the technical realization of the system, documenting the specific technologies utilized and the validation measures undertaken. It moves beyond the design abstractions of Chapter 3 to present the actual software artifacts, including the user interfaces, the integrated development environment, and the rigorous testing methodologies used to verify system correctness before deployment at MTN Rwanda.

The transition from design to construction was guided by the UML artifacts created in the previous phase. The class diagrams provided the direct blueprint for the Mongoose models, ensuring that every data field required for productivity analytics was correctly mapped in the database. Similarly, the sequence diagrams acted as a step-by-step logic guide for developing the Express.js controllers and the AI classification middleware. By strictly adhering to these diagrams, the implementation maintained a high degree of architectural integrity, ensuring that the final software reflected the original requirements. This alignment was particularly important when integrating the TensorFlow.js model, as it required precise data hand-offs between the desktop monitoring service and the analytical backend to ensure that classification happened in near real-time.

Furthermore, this phase prioritized the creation of a responsive and intuitive user experience by translating the wireframes into dynamic React components that reflect real-time data updates. This way the transmission of data is near instantaneous

Technologies and Tools used

The ProTrackAI system was built using a modern technology stack carefully selected to meet the requirements of scalable activity tracking, intelligent classification, and responsive user interfaces. This section details the specific technologies, programming languages, frameworks, and tools employed across the frontend, backend, database, and development infrastructure.

Programming Languages

JavaScript (ES6+):

JavaScript served as the primary programming language for both client-side and server-side development, enabling code consistency and knowledge reuse across the entire application stack. On the frontend, JavaScript powered the React components and user interface logic within web browsers. On the backend, JavaScript ran on Node.js runtime to handle HTTP requests, database operations, and AI classification processing. The use of modern ECMAScript 2015 and later features such as arrow functions, destructuring, async/await syntax, template literals, and module imports improved code readability and maintainability throughout the codebase.

Frontend Technologies

React (Version 18):

React served as the foundation for building the web-based user interface. This JavaScript library enables creation of reusable UI components that efficiently update when underlying data changes through its virtual DOM reconciliation algorithm. I organized the ProTrackAI frontend into functional components representing distinct interface sections such as dashboards, charts, forms, and navigation menus. React's component-based architecture allowed me to build complex interfaces by composing smaller, focused components that maintain their own state and lifecycle methods. The use of React Hooks including `useState` for component state management, `useEffect` for side effects and data fetching, `useContext` for global authentication state, and custom hooks for shared logic patterns simplified component code and promoted reusability.

Material UI (Version 6):

Material UI is a comprehensive React component library implementing Google's Material Design specifications. This library provided pre-built, customizable components including buttons, text fields, cards, tables, grids, navigation drawers, dialogs, and icons that accelerated

interface development significantly. Material UI components come with built-in accessibility features, responsive behavior, and theme customization capabilities. I configured a custom theme defining the MTN Rwanda color palette with primary yellow as the accent color and secondary black for contrast, typography settings using modern sans-serif fonts for improved readability, spacing units for consistent padding and margins across components, and breakpoints defining how layouts adapt to different screen sizes from mobile phones to desktop monitors.

Vite:

Vite served as the build tool and development server for the frontend application. Vite offers significantly faster development server startup times compared to traditional bundlers by serving source files directly over native ES modules in the browser during development. When building for production, Vite leverages Rollup to create optimized, minified JavaScript bundles with efficient code splitting. The hot module replacement feature allowed me to see code changes reflected in the browser instantly without full page reloads, dramatically improving development productivity.

Chart.js and react-chartjs-2:

Chart.js is a flexible JavaScript charting library for creating interactive data visualizations. The react-chartjs-2 wrapper integrates Chart.js seamlessly with React components. I used these libraries to implement the doughnut charts displaying productivity time distribution on employee dashboards, bar charts showing team member performance comparisons on supervisor dashboards, and line charts illustrating productivity trends over time. The charts respond to user interactions such as hovering to display detailed tooltips and clicking legend items to toggle data series visibility.

Axios:

Axios is a promise-based HTTP client for making API requests from the frontend to the backend. Axios simplifies request configuration by providing a clean API for setting headers, handling request and response interceptors, managing timeout configurations, and automatically transforming JSON data. I configured Axios with a base URL pointing to the backend API server, default headers including the authorization token for authenticated requests, and interceptors to handle token expiration by redirecting users to the login page when receiving unauthorized responses.

Socket.IO Client:

The Socket.IO client library enables real-time bidirectional communication between the frontend and backend over WebSocket connections. I used Socket.IO to implement the live notification system that delivers alerts to users without requiring page refresh. When users log in, the client establishes a persistent connection to the server and joins rooms corresponding to their role and team. The client listens for notification events and updates the notification badge counter and dropdown list dynamically when new alerts arrive.

Backend Technologies**Node.js:**

Node.js provided the runtime environment for executing JavaScript code on the server. Built on Chrome's V8 JavaScript engine, Node.js uses an event-driven, non-blocking I/O model that makes it efficient for handling concurrent requests typical in web applications. The asynchronous nature of Node.js allowed the backend to process multiple activity ingestion requests simultaneously without blocking threads while waiting for database operations to complete. Node.js also provided access to the npm package ecosystem containing thousands of libraries for various functionalities.

Express.js:

Express is a minimal and flexible web application framework for Node.js that provides robust features for building APIs and web applications. Express simplified the creation of RESTful API endpoints by offering intuitive routing mechanisms to map HTTP methods and URL patterns to handler functions, middleware support for implementing cross-cutting concerns like authentication and logging, request and response objects with convenient methods for parsing input and sending output, and error handling middleware for centralized exception management. The fourteen route modules I created for authentication, users, activities, reports, tasks, goals, feedback, evaluations, time tracking, analytics, notifications, AI configuration, leave management, and organizational structure all built upon Express routing capabilities.

TensorFlow.js:

TensorFlow.js is a JavaScript library for training and deploying machine learning models in Node.js and browser environments. I used TensorFlow.js to build the neural network classifier

that serves as the final classification layer when rule-based logic produces low confidence results. The library provided APIs for constructing neural network architectures using layers, training models on labeled datasets using backpropagation, saving trained models to disk for persistence, loading models into memory for inference, and making predictions on new input data. The classifier uses a sequential model with embedding, flatten, dense, and dropout layers to transform activity text into productivity categories.

Mongoose (Version 7.x):

Mongoose is an Object Document Mapper for MongoDB and Node.js that provides a schema-based solution for modeling application data. Mongoose offered significant benefits over raw MongoDB drivers including schema definition with type validation ensuring data consistency, relationships through population of referenced documents, middleware hooks for executing code before or after database operations like hashing passwords before save, query builder for constructing complex database queries, and promise support for asynchronous operations. I defined ten Mongoose schemas corresponding to the collections for users, activities, time logs, goals, tasks, feedback, evaluations, notifications, leave requests, and organizational structure.

Socket.IO Server:

The Socket.IO server library enabled real-time communication by maintaining persistent WebSocket connections with connected clients. Socket.IO handles the complexity of establishing connections, managing fallbacks to HTTP long-polling when WebSockets are unavailable, broadcasting messages to multiple clients, and organizing clients into rooms for targeted message delivery. I configured Socket.IO to integrate with the Express server, allowing it to handle WebSocket upgrade requests on the same port as HTTP traffic. The server emits notification events to specific rooms when backend operations create notifications, ensuring alerts reach appropriate users efficiently.

PDFKit:

PDFKit is a JavaScript library for generating PDF documents programmatically. I used PDFKit to create professional productivity reports that users can download for record-keeping or presentation purposes. The library allows precise control over document layout including page size and margin settings, text positioning with font selection and styling, drawing shapes and

lines for visual separators, embedding images for branding elements, and creating tables for structured data presentation. The PDF generation functions stream binary data directly to HTTP responses rather than writing temporary files, improving efficiency and security.

JSON Web Token (JWT):

JWT is a compact, URL-safe means of representing claims to be transferred between parties. I implemented JWT-based authentication where users receive a token upon successful login. This token contains encoded user information including identifier, role, team, and expiration time. Clients include this token in the Authorization header of subsequent API requests. The backend middleware verifies the token signature using a secret key, ensuring it has not been tampered with, and extracts the user information for authorization checks. The stateless nature of JWT eliminates the need for server-side session storage, simplifying horizontal scaling of the backend.

bcrypt:

bcrypt is a password hashing library designed specifically for securing passwords. Unlike fast hashing algorithms like MD5 or SHA, bcrypt is intentionally slow and includes a salt to protect against rainbow table attacks. When users register or change passwords, the backend hashes the password using bcrypt before storing it in the database. During login, the provided password is hashed and compared to the stored hash using bcrypt's time-constant comparison to prevent timing attacks. This approach ensures that even if the database is compromised, attackers cannot easily recover plaintext passwords.

Database Technologies

MongoDB:

MongoDB served as the primary database for storing all application data. This NoSQL document database stores data in flexible JSON-like documents rather than rigid table structures. MongoDB's schema flexibility proved valuable for handling evolving activity metadata structures that varied based on application types and usage contexts. The database supports rich queries including field matching, range queries, regular expression searches, and aggregation pipelines for complex analytics computations. MongoDB's horizontal scaling capabilities through sharding provide a path for growth as activity volumes increase. I organized data into ten collections representing core domain entities with strategic indexing on frequently queried fields to optimize performance.

Mongoose ODM:

As mentioned in the backend technologies section, Mongoose provided the object modeling layer on top of MongoDB. Mongoose schemas defined the structure and validation rules for documents in each collection, ensuring data consistency despite MongoDB's flexible nature. Features like virtual properties for computed fields, instance methods for document-specific operations, and static methods for collection-level operations enhanced the expressiveness of database interactions throughout the codebase.

Development Tools and Infrastructure

Visual Studio Code:

Visual Studio Code served as the integrated development environment for writing, debugging, and managing code. This lightweight, extensible editor provided syntax highlighting for JavaScript, HTML, and CSS, intelligent code completion through IntelliSense, integrated terminal for running commands, debugging capabilities for both Node.js and browser JavaScript, Git integration for version control operations, and a rich extension ecosystem including ESLint for code quality checks and Prettier for automatic code formatting.

Git:

Git provided distributed version control for tracking code changes throughout development. I initialized Git repositories for both the backend and frontend projects, committing changes regularly with descriptive messages documenting modifications. Git enabled experimentation with new features on separate branches without affecting the stable main codebase, allowing me to merge successful implementations while discarding failed approaches. The commit history served as a development diary documenting the evolution of the system over time.

npm (Node Package Manager):

npm managed JavaScript dependencies for both frontend and backend projects. The package.json files in each project specified required libraries with version constraints. Running npm install downloaded all dependencies and their transitive dependencies into the node_modules directory. The package-lock.json files locked exact versions of all packages, ensuring consistent dependency trees across different development machines and production servers. npm scripts

defined in `package.json` provided convenient commands for starting development servers, building production bundles, and running tests.

Postman:

Postman is an API development tool that I used extensively for testing backend endpoints during development. Postman allowed me to construct HTTP requests with specific methods, headers, and body content, send requests to the backend API, examine detailed responses including status codes and headers, and organize requests into collections representing different features. I created a ProTrackAI collection with example requests for all API endpoints, facilitating rapid testing of new functionality and debugging of issues without needing to build frontend interfaces first.

Electron:

Electron is a framework for building cross-platform desktop applications using web technologies. The ProTrackAI desktop agent was built with Electron, allowing it to run on Windows, macOS, and Linux systems despite being written in JavaScript. Electron provided access to native operating system APIs for capturing active window information, created system tray icons for minimized application access, implemented custom protocol handlers for deep linking authentication tokens, and packaged the application as standalone executables for distribution. The Electron main process managed application lifecycle and system integration while renderer processes handled the minimal user interface for the login window.

Presentation of the New System

Login Interface

The entry point to the system, focusing on security and brand identity. Users authenticate via email or employee ID.

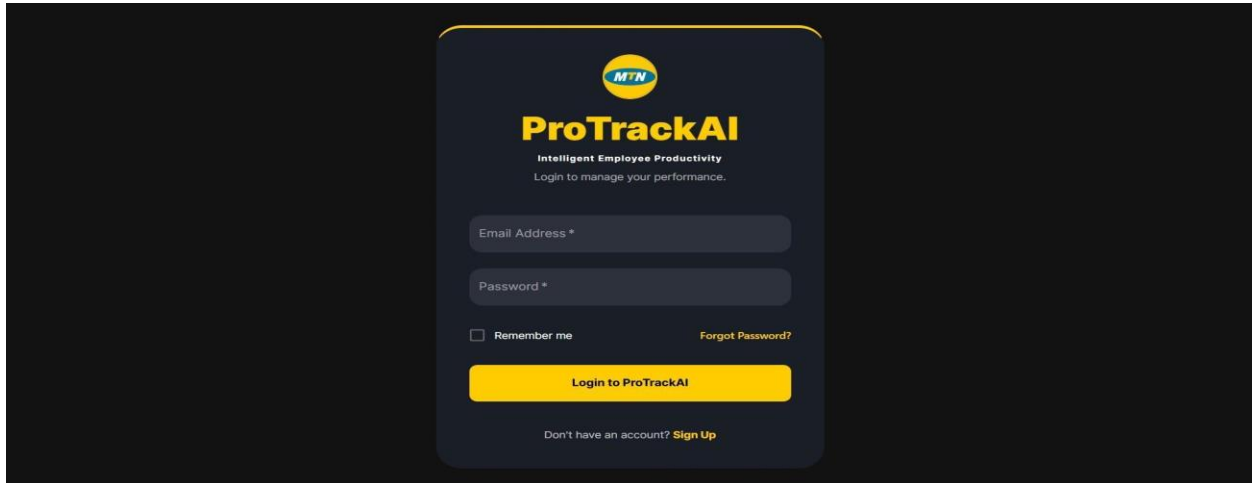
The login interface for ProTrackAI is displayed on a dark background. At the top, the MTN logo is shown above the text "ProTrackAI" and "Intelligent Employee Productivity". Below this, there are input fields for "Email Address *" and "Password *". A "Remember me" checkbox and a "Forgot Password?" link are positioned below the password field. A prominent yellow button labeled "Login to ProTrackAI" is centered below the inputs. At the bottom, a link for "Don't have an account? Sign Up" is visible.

Figure 8: Login Interface

Registration

A streamlined form for new employee onboarding, capturing essential personal and departmental details.

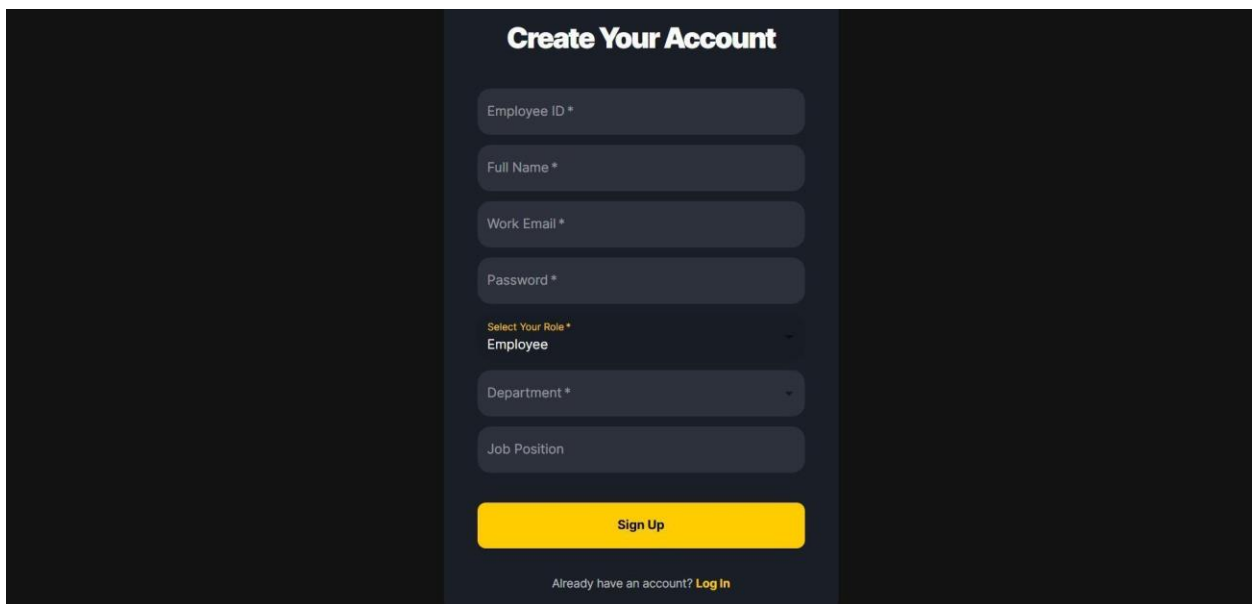
The registration interface, titled "Create Your Account", is shown on a dark background. It features a vertical stack of input fields: "Employee ID *", "Full Name *", "Work Email *", "Password *", "Select Your Role *" (with "Employee" selected), "Department *", and "Job Position". A yellow "Sign Up" button is located at the bottom of the form. Below the button, a link for "Already have an account? Log In" is provided.

Figure 9: Registration Interface

Employee Dashboard

The central hub displaying real-time clock-in status, daily productivity scores, and recent activity insights

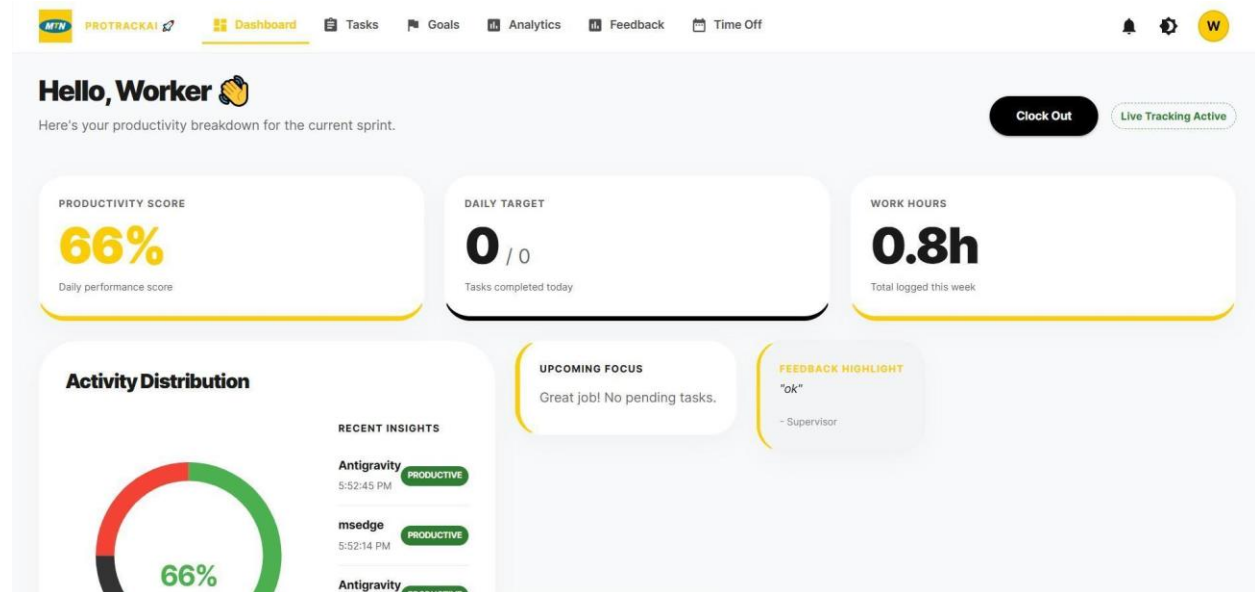


Figure 10: Employee dashboard

Task Board

Allows employees to view assigned tasks, track deadlines, and update status (Pending, In Progress, Completed)

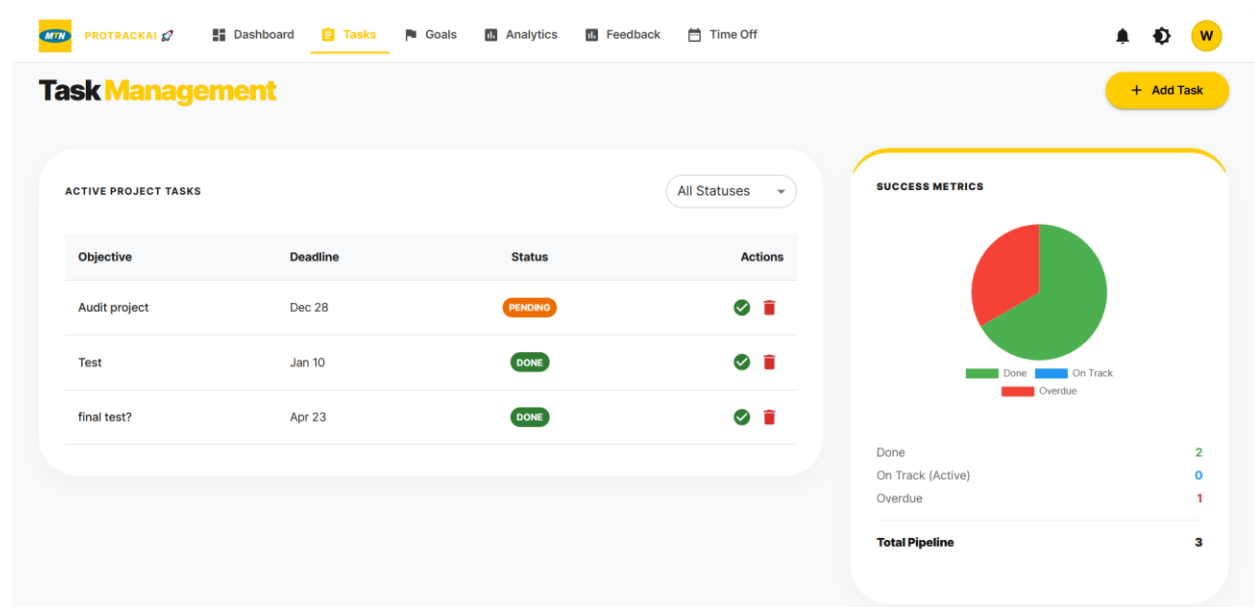


Figure 11: Task Management

Goal Tracker

Visualizes progress towards strategic objectives. Employees can update completion percentages and view supervisor feedback.

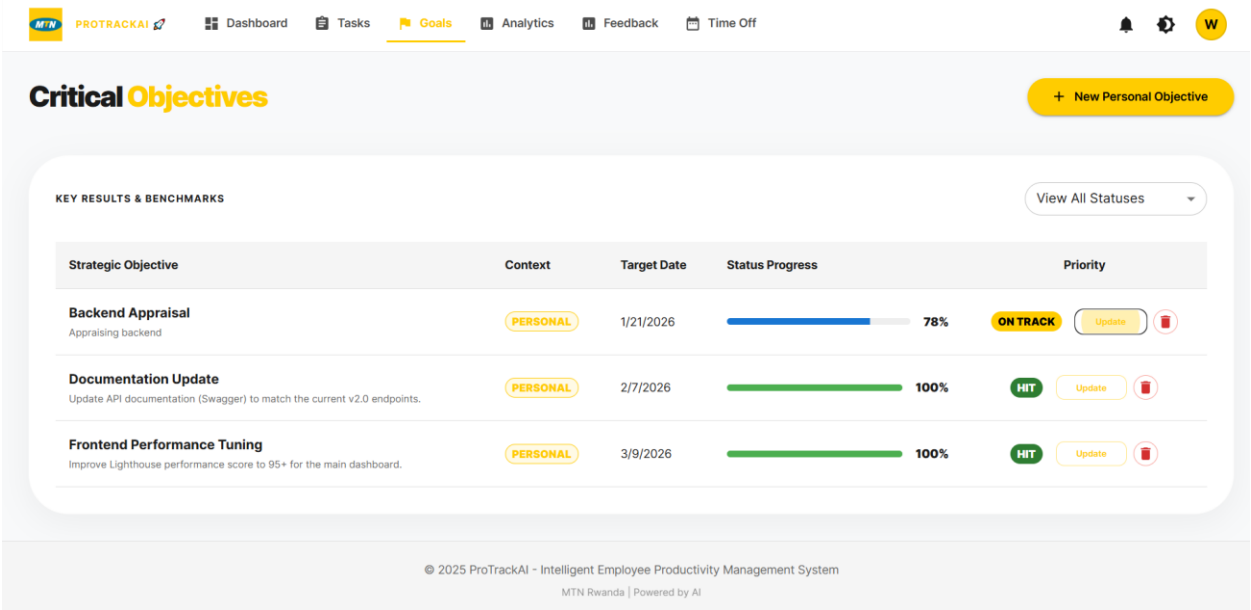


Figure 12: Goal Management Interface

Productivity Reports

Detailed analytics view allowing employees to analyze their own work habits over daily, weekly, or monthly periods

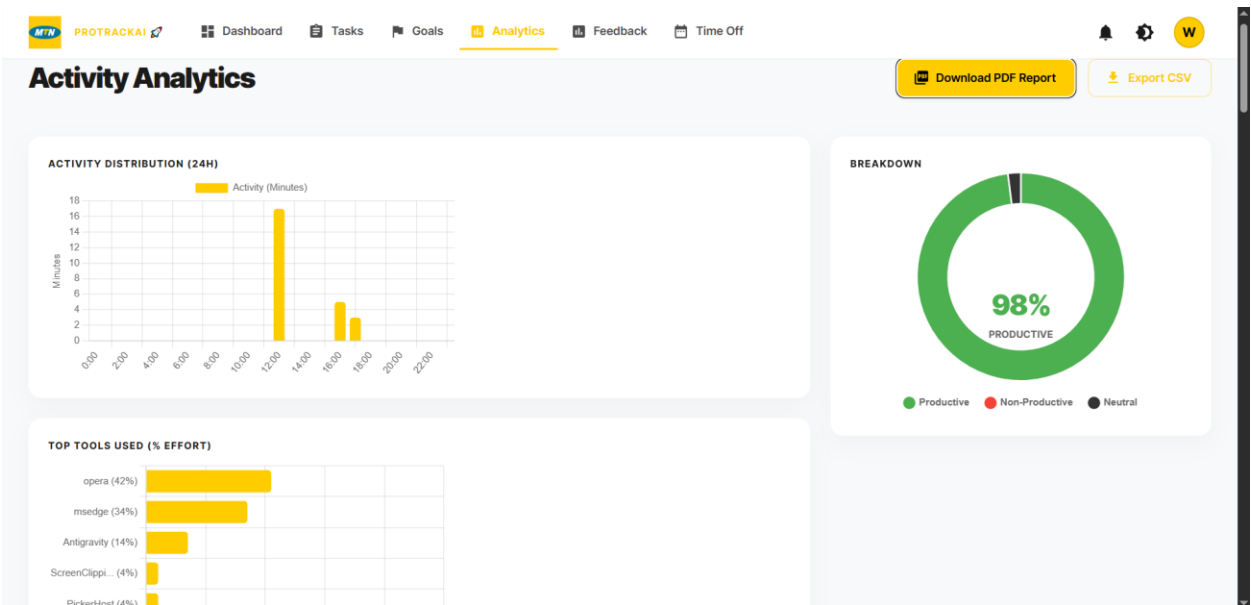


Figure 13: Personal Productivity Report

Activity Timeline

A chronological log of all captured system activities, showing the AI's classification for each action

Antigravity 1/8/2026, 4:07:44 PM - 30s	productive
opera 1/8/2026, 4:07:12 PM - 30s	productive
opera 1/8/2026, 4:06:41 PM - 30s	productive
opera 1/8/2026, 4:06:10 PM - 30s	productive
msedge 1/8/2026, 4:05:37 PM - 30s	productive
msedge 1/8/2026, 4:05:06 PM - 30s	productive
opera 1/8/2026, 4:04:02 PM - 30s	productive
opera 1/8/2026, 4:03:29 PM - 30s	productive
opera 1/8/2026, 4:02:55 PM - 30s	productive

Figure 14: Detailed Activity Log

Leave Management

Interface for submitting leave requests (Annual, Sick, Emergency) and tracking approval status from HR.

PROTRACKAI

Dashboard

Tasks

Goals

Analytics

Feedback

Time Off

3

A

LeaveManagement

+ Request Time Off

MY REQUESTS HISTORY

Type	Dates	Reason	Proof	Status
BEREAVEMENT	3/19/2026 - 4/11/2026 Duration: 17 Work Days	speed ticket	View Note	PENDING
VACATION	3/27/2026 - 3/25/2026 Duration: 0 Work Days	sick	--	PENDING
OTHER	2/22/2026 - 2/22/2026 Duration: 0 Work Days	Family emergency/Personal matter.	--	PENDING
SICK	2/16/2026 - 2/16/2026 Duration: 1 Work Days	Flu symptoms.	--	APPROVED
VACATION	2/27/2026 - 3/4/2026 Duration: 4 Work Days	Annual family vacation to Zanzibar.	--	APPROVED

Figure 15: Leave management

Notifications Center

Real-time alerts for new tasks, goal updates, and system announcements.

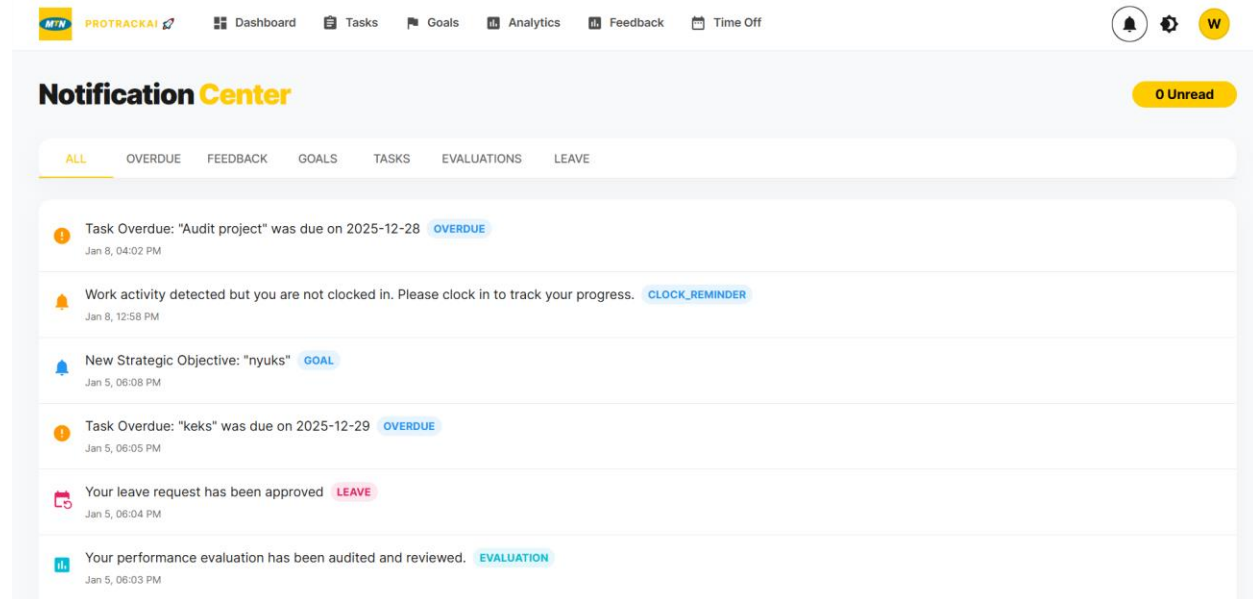


Figure 16: Notifications Panel

User Profile

Settings page for updating personal information, changing passwords, and uploading a profile picture.

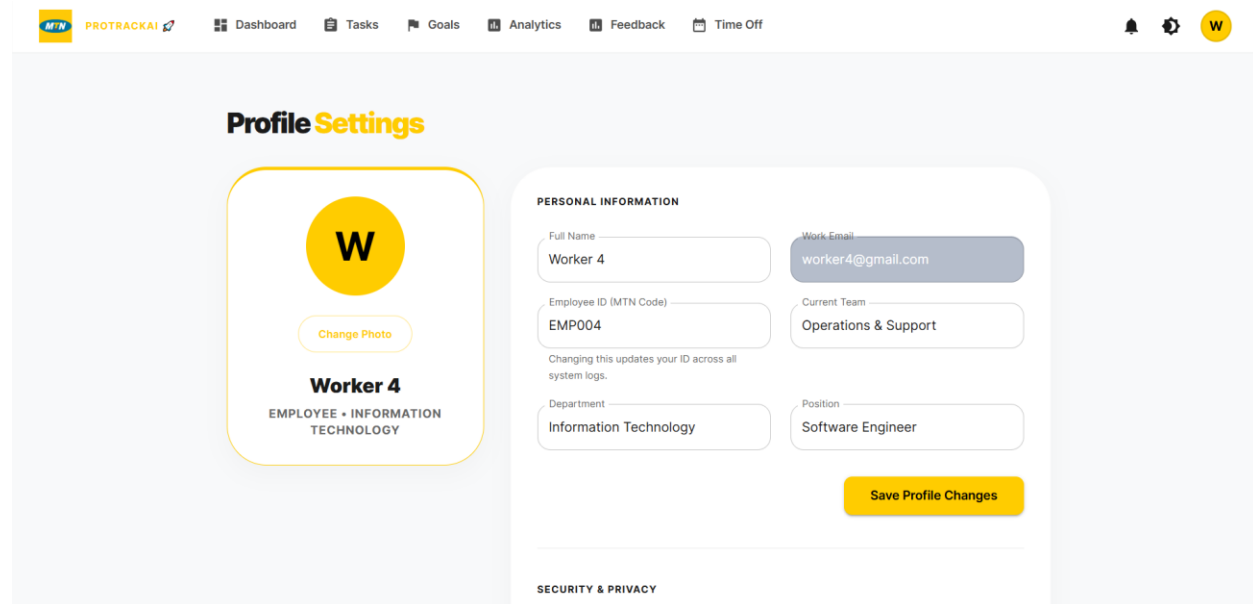


Figure 17::Profile Settings

Supervisor Dashboard

Command center for team oversight, displaying aggregate team performance metrics and critical alerts.

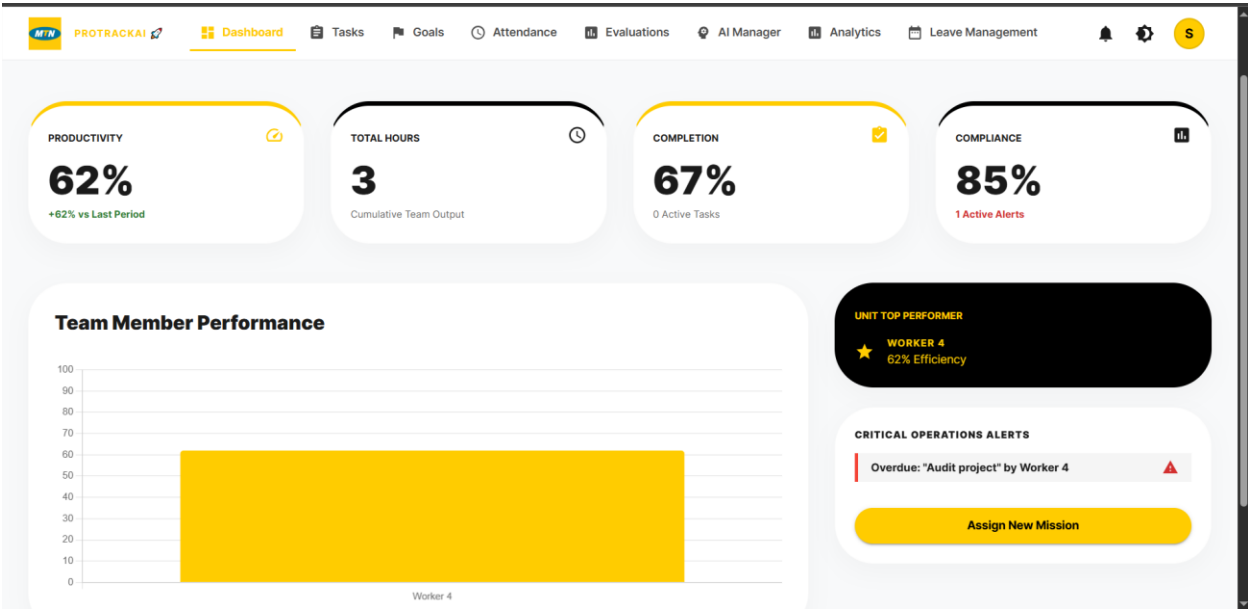


Figure 18: Supervisor dashboard and Team members performance

Goal Assignment

Tools for supervisors to create new strategic goals and assign them to specific team members.

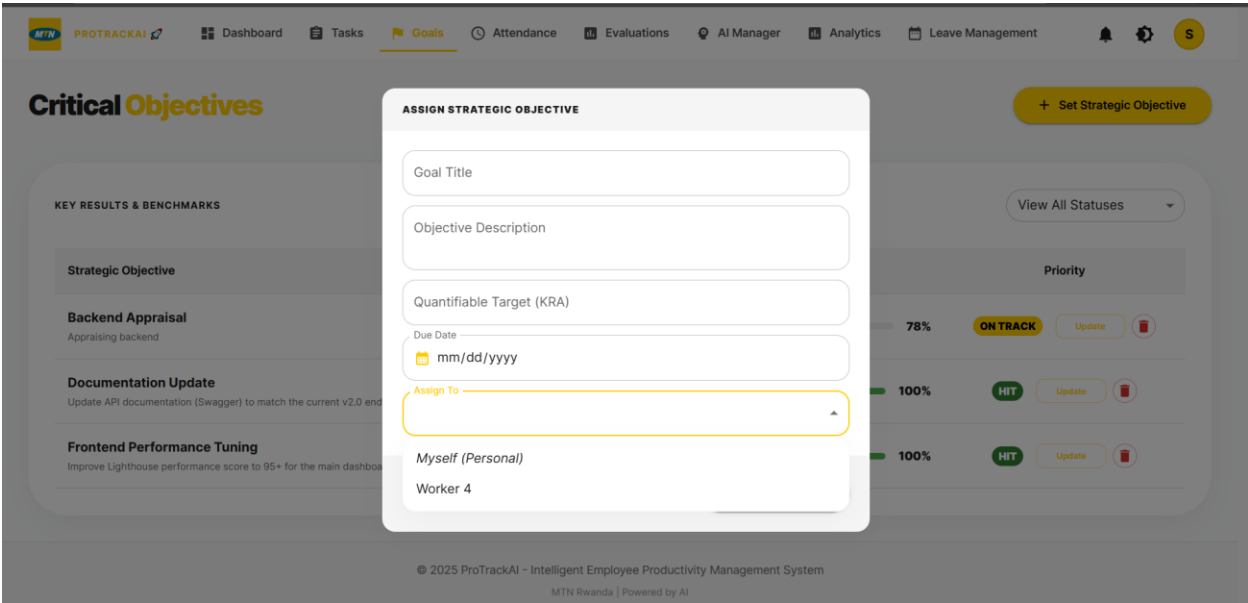


Figure 19: Goal Assignment Form

AI Classifier Tester

A feature for a supervisor to test the effectiveness of the classifying model

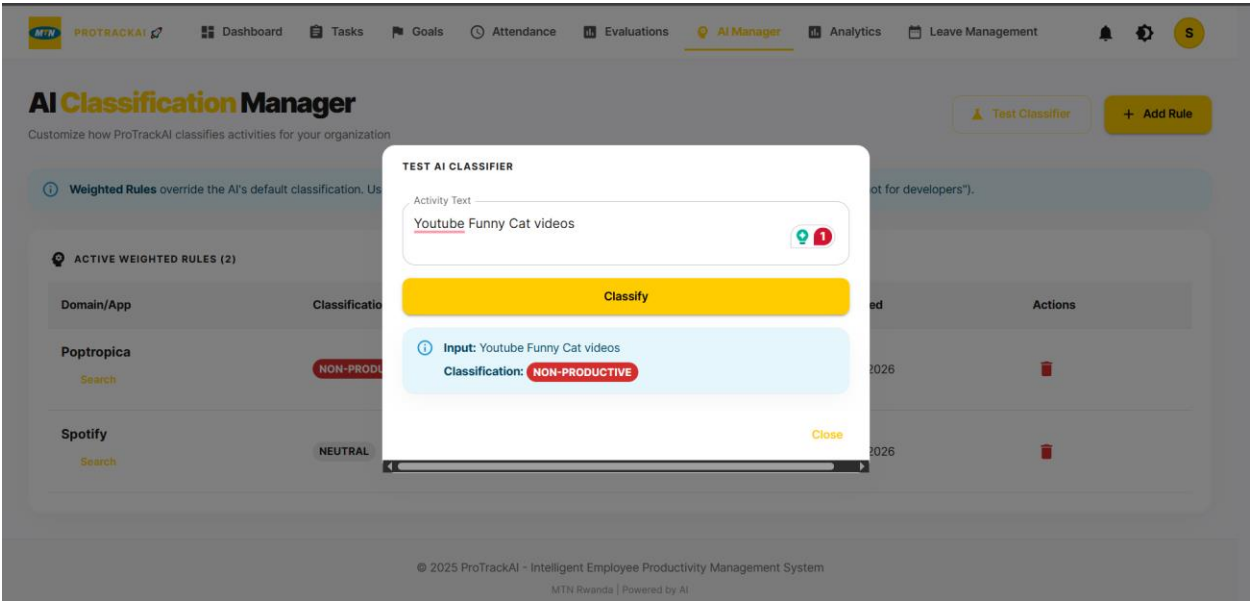


Figure 20: AI classifier tester

AI Weighted Rule Configuration

Allows supervisors to define custom productivity rules (e.g., specific URLs or Apps) for their department.

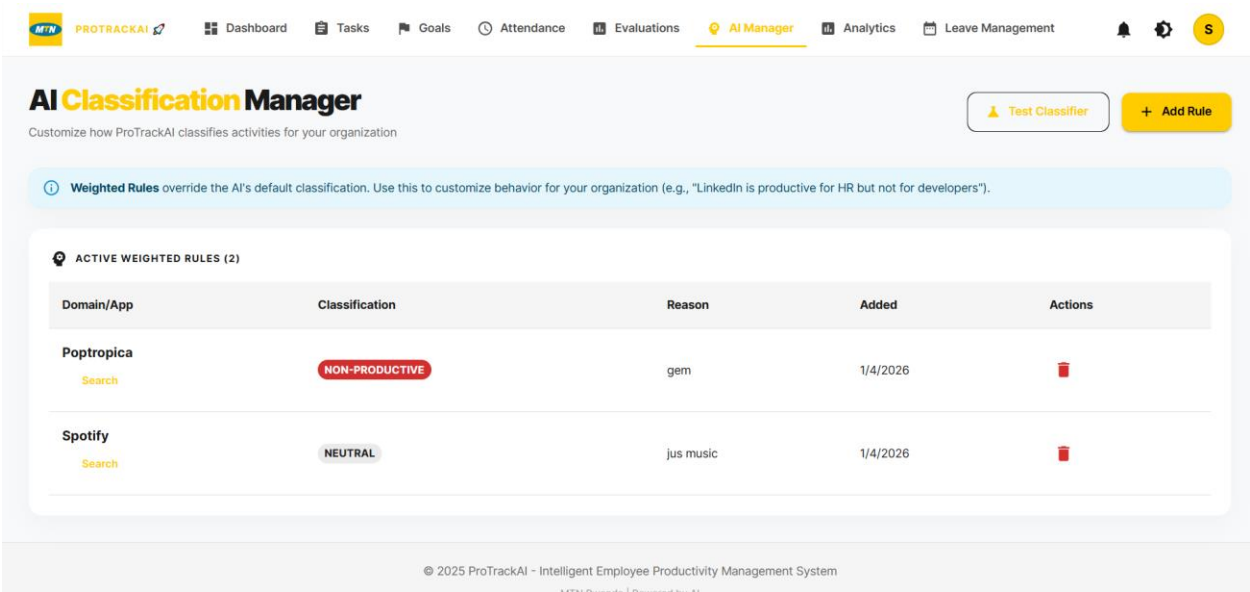


Figure 21::Weighted rules

Hr Dashboard

Command center for overseeing departments and teams and the general productivity rate of them all with more detail and trend display as well as what needs attention

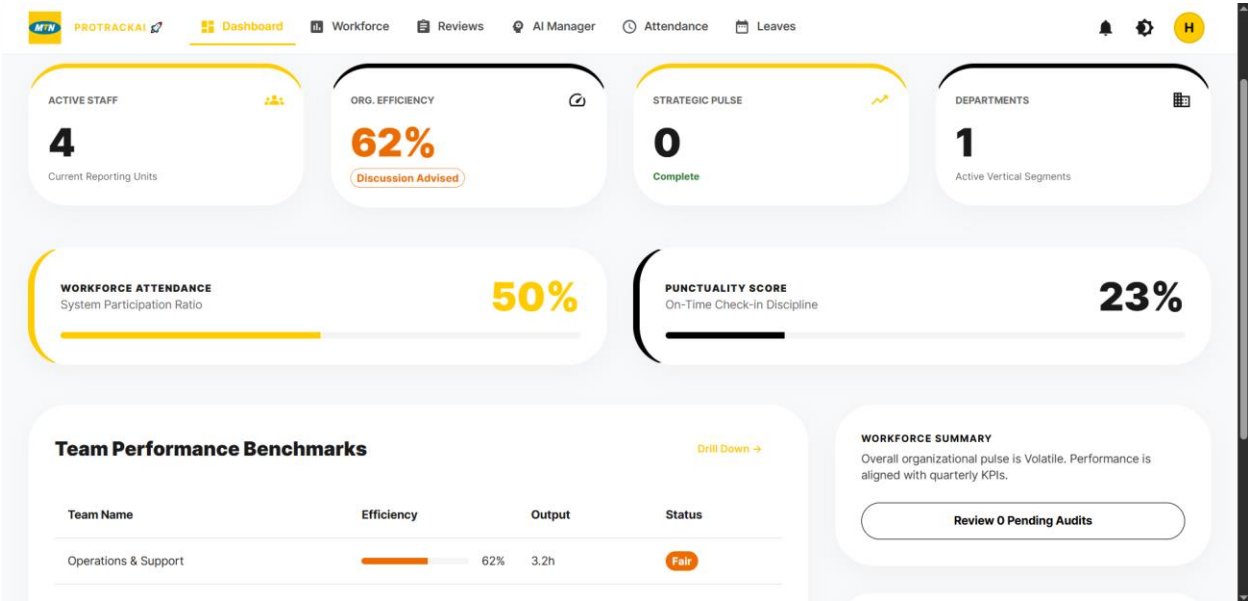


Figure 22::Hr Dashboard

Department Management

Where the HR manages Departments and functional teams within that department whether it be addition, deletion or renaming

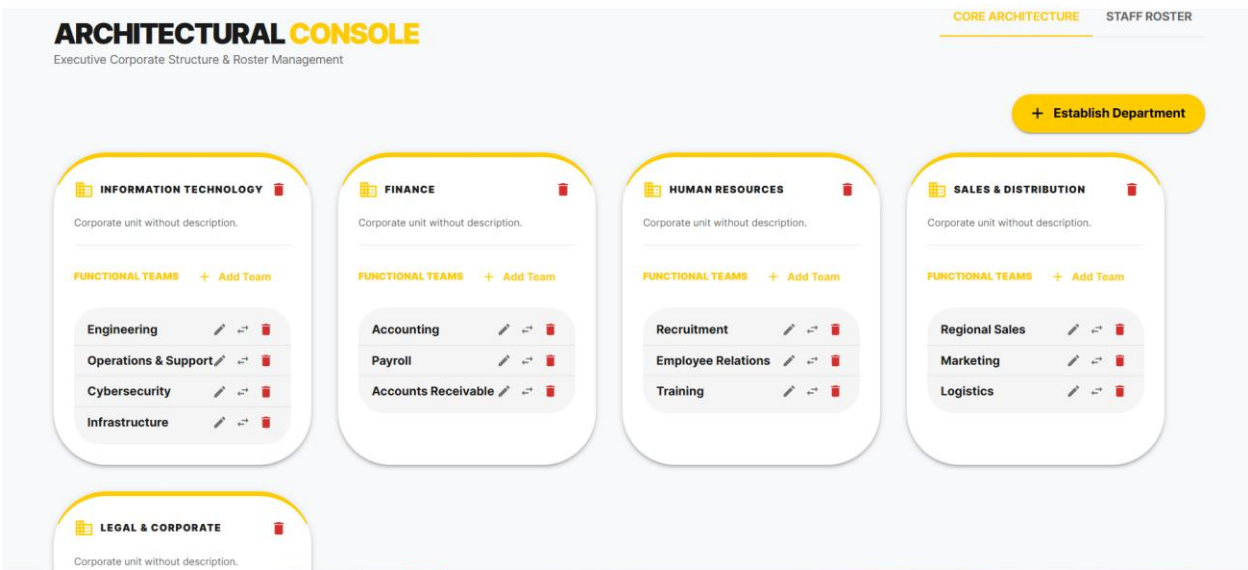
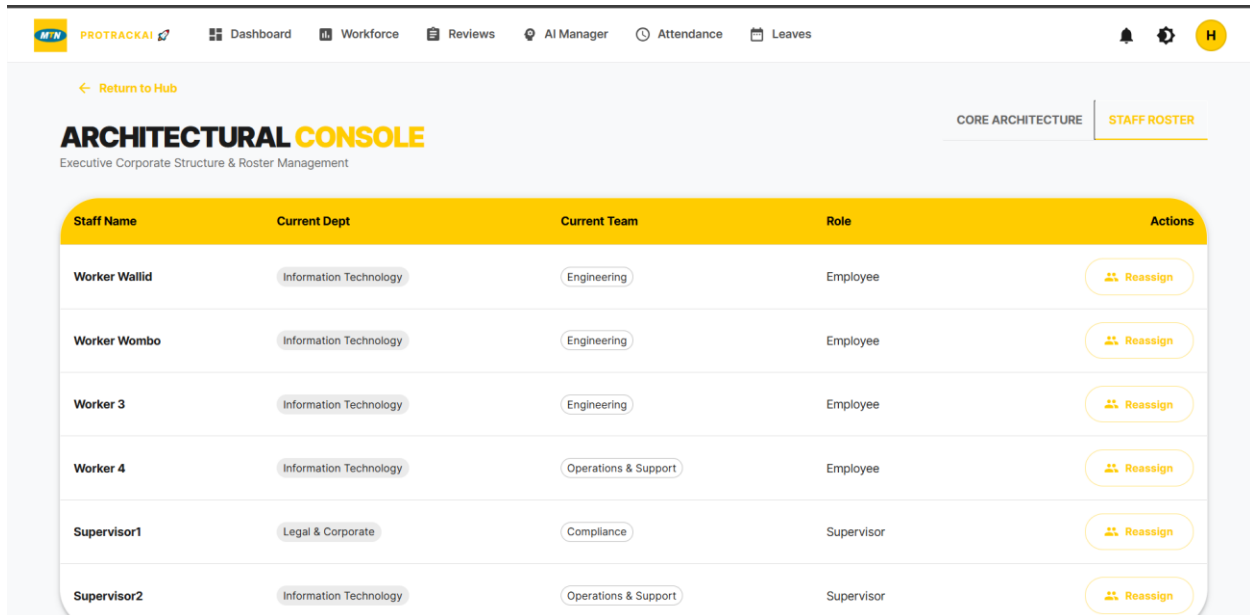


Figure 23: Department Management

Team Assignment

Where teams assignments happen, when the HR needs to switch employees or supervisors from one department/team to another



The screenshot shows the PROTRACKAI interface with the 'STAFF ROSTER' tab selected. The table lists staff members with their current department, team, and role, along with a 'Reassign' button for each.

Staff Name	Current Dept	Current Team	Role	Actions
Worker Wallid	Information Technology	Engineering	Employee	<button>Reassign</button>
Worker Wombo	Information Technology	Engineering	Employee	<button>Reassign</button>
Worker 3	Information Technology	Engineering	Employee	<button>Reassign</button>
Worker 4	Information Technology	Operations & Support	Employee	<button>Reassign</button>
Supervisor1	Legal & Corporate	Compliance	Supervisor	<button>Reassign</button>
Supervisor2	Information Technology	Operations & Support	Supervisor	<button>Reassign</button>

Figure 24: Team/Department worker assignment

Work Force Intelligence

Where the tools that were most detrimental and most useful to the departments are displayed for intelligence purposes so that decisions may be made to aid productivity

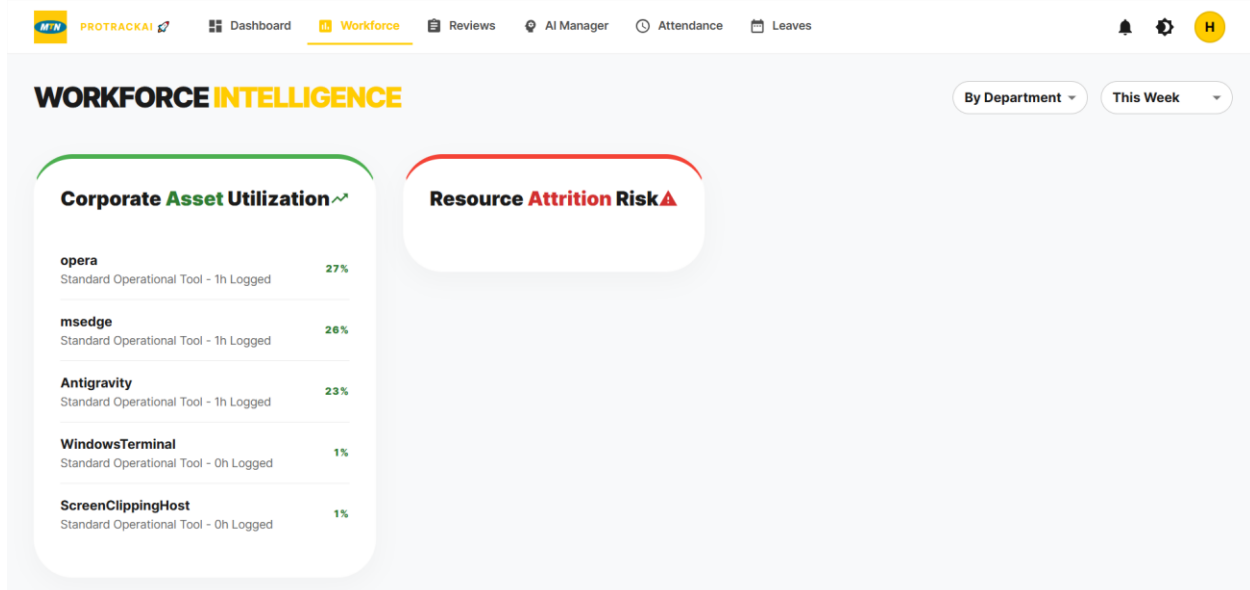


Figure 25: Work Force intelligence

Report Generation

Reports of varying clearance levels (Hr, Supervisor, employees) can be downloaded in pdf format on each dashboard to convey productivity trends according to who requests it

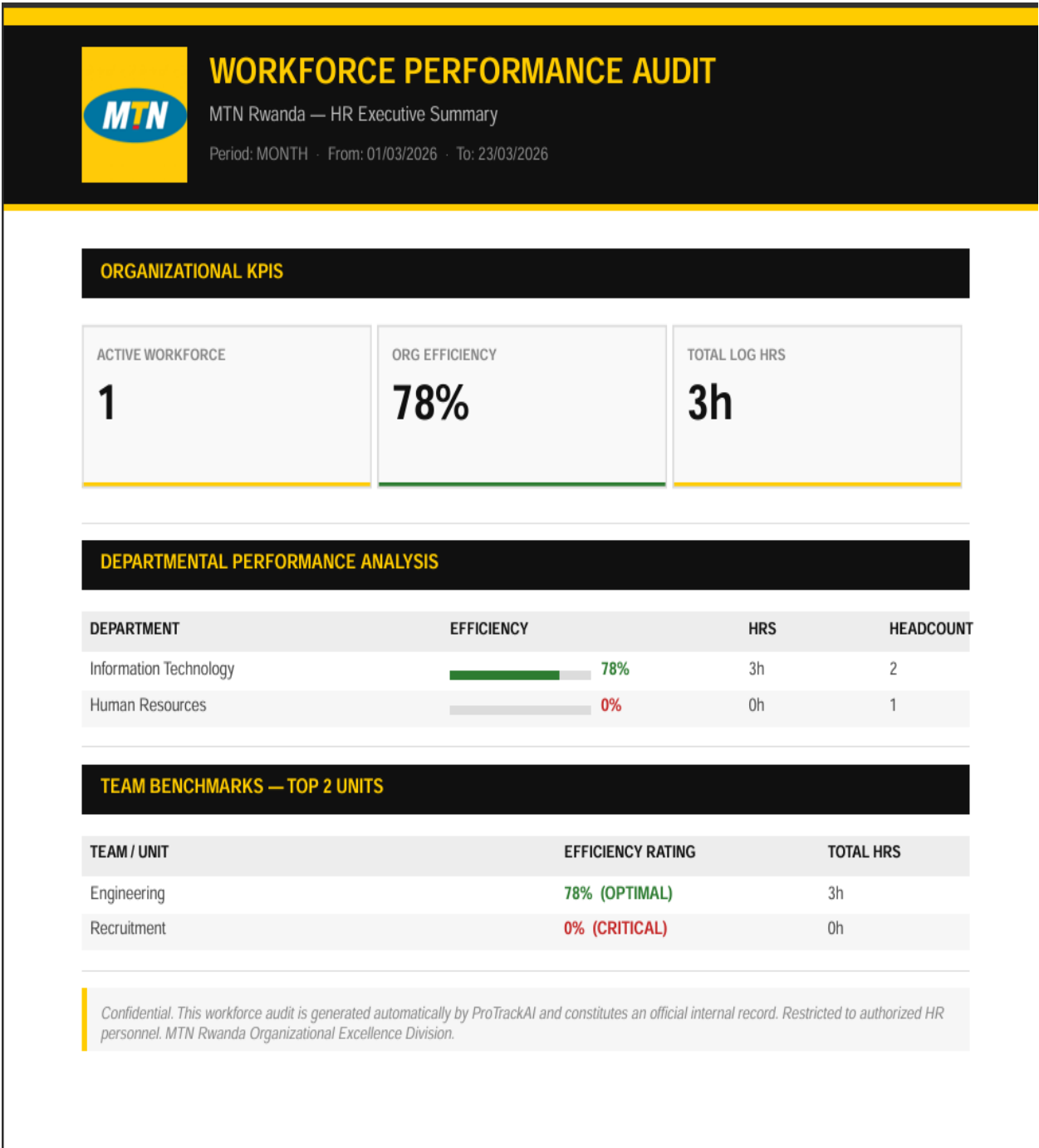


Figure 26: Report Generation

Software Testing

To ensure the reliability and accuracy of the ProTrackAI system, a rigorous testing phase was conducted throughout the development lifecycle. This process focused on validating that all functional requirements, particularly the AI-driven activity classification, operated as intended under various conditions.

The primary objectives of the testing strategy were to:

- **Guarantee Data Accuracy:** Ensuring that employee work logs and productivity scores are calculated precisely.
- **Validate AI Performance:** Confirming that the machine learning model correctly distinguishes between productive and non-productive activities.
- **Secure User Data:** Verifying that robust access controls protect sensitive organizational data.

The following testing methodologies were employed:

Unit Testing: Ensuring individual modules like the AI text classifier, JWT authentication utility, and React components function correctly in isolation. For example, verifying that the `calculateProductivity` function returns correct percentages for known inputs.

Integration Testing: Checking that separate modules such as the desktop agent, backend API, and MongoDB database create a seamless data pipeline. This ensured that activity data sent from the desktop client was correctly parsed, classified, and stored in the database.

System Testing: Verifying end-to-end workflows from user perspective. This included testing the complete journey of a new employee registering, installing the agent, tracking work, and viewing their daily report.

Security Testing: Ensuring data protection for sensitive employee activity logs and personal information. Tests verified that API endpoints were properly protected by JWT authentication and that users could not access data belonging to other organizations or teams.

User Acceptance Testing (UAT): Validating that the system met the core business requirements of MTN Rwanda, specifically the ability to distinguish between "Productive" and "Non-Productive" work hours without manual intervention.

Hardware and Software Requirements

This section outlines the hardware and software requirements for operating the ProTrackAI system efficiently.

Client-Side Requirements (Employee Workstations)

- Operating System: Windows 11, macOS (Catalina+), or Linux (Ubuntu 20.04+).
- Web Browser: Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari (latest versions).
- Hardware: Minimum 16GB RAM, 2.0 GHz Dual-Core Processor, and stable internet connection.

Server-Side Requirements (Backend Infrastructure)

- Database: MongoDB 6.0+ (Community or Enterprise).
- Runtime: Node.js v18.0+.
- Hardware: Recommended 4GB+ RAM and 2 vCPU cores for small to medium deployments.
- Network: HTTPS support (SSL/TLS) for secure data transmission.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

Conclusion

The ProTrackAI project was initiated to address a critical gap identified at MTN Rwanda: the inability to accurately measure and optimize employee productivity in a digital work environment. While the organization possessed basic internet logging capabilities, these tools provided only passive data without context, failing to distinguish between productive research and non-productive distractions, hindering management's ability to make data-driven decisions.

To resolve this, ProTrackAI was developed as an intelligent, active monitoring solution. By leveraging a custom-trained AI classifier built on a MERN stack architecture, the system automates the categorization of digital activities, providing real-time visibility into workforce performance while respecting user privacy through domain-level reporting and transparent personal dashboards. The integration of Management by Objectives features further allowed productivity to be contextualized against assigned strategic goals rather than raw time metrics alone.

Ultimately, ProTrackAI transforms workplace monitoring from surveillance to empowerment, shifting the focus from time-tracking to measuring actual engagement and output quality. By providing employees with access to their own productivity analytics, the system fosters a culture of self-regulation and continuous improvement. This research successfully delivered a scalable, privacy-conscious productivity management tool that validates the premise that AI-driven insights, when applied ethically, can significantly enhance both organizational efficiency and employee satisfaction.

Recommendations

Based on the development findings and pilot observations, the following recommendations are proposed for successful implementation and future enhancement.

to MTN Rwanda

- **Phased Pilot Deployment:** Rather than an immediate organization-wide rollout, implementation should begin with a single technical department. This allows for the refinement of AI classification rules specific to MTN's internal tools before scaling to the broader workforce.
- **Transparency-First Policy:** To build trust, management must clearly communicate that the system measures productivity trends, not personal surveillance. Policies should explicitly state that private message content and passwords are never recorded.
- **Development-Focused Usage:** Productivity scores should be used primarily for identifying training needs and resource gaps, rather than punitive measures. High performers identified by the system should be recognized to incentivize adoption.

To Future Researchers

- **Mobile Application Extension:** As remote work increases, future iterations should include a mobile companion app. This would allow field employees (e.g., Sales agents) to log activities and view their stats on the go, independent of the desktop agent.
- **Predictive Burnout Modeling:** Advanced machine learning models could be integrated to analyze prolonged periods of high intensity or late-night activity, sending proactive alerts to supervisors to prevent employee burnout before it occurs.
- **Cross-Platform Native Agents:** While the Electron agent works well for Windows, developing native agents for macOS and Linux using languages like Rust or Swift would further optimize resource usage and system integration for diverse technical teams.

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APPENDICES

Curriculum Vitae

JESSE SHEMA KYANGA

*Software Engineer |
Backend & Full-Stack
Development*

PROFILE

Software Engineer with strong foundations in backend and full-stack development, experienced in Java, Spring Boot, React, SQL, and Docker. Completed all academic coursework and currently in the final defense stage. Proven ability to build, deploy, and maintain real-world systems with clean architecture and good engineering practices.

CONTACT ME

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✉️ jessbra4@gmail.com

🐙 <https://github.com/Jesse-Shema-Kyanga>

➤ EDUCATION

ADVENTIST UNIVERSITY OF CENTRAL AFRICA 2022 – 2026

BSc Software Engineering
All coursework completed · Final defense pending

➤ LANGUAGE

Native Kinyarwanda
Advanced English
Advanced French

➤ SKILLS

Languages: Python, Java, JavaScript, SQL

Frameworks: Spring Boot, React

Expertise: Backend Development, REST APIs, Data & Machine Learning

Tools: Docker, Git, Linux

Other: Cybersecurity fundamentals

➤ EXPERIENCE

SESMEC LTD

Data Transcriptionist

Transcribed and structured survey data for analysis.

MTN RWANDA – TECHNOLOGY

DEPARTMENT

Software Engineering Intern

Gained exposure to production environments and system workflows

Data Collection Letter



Adventist University of Central Africa

P.O. Box 2461 Kigali, Rwanda | www.auca.ac.rw | info@auca.ac.rw

Faculty of Information Technology

July 09, 2025

TO: MTN

Dear Sir/Madam,

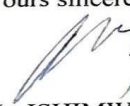
Re: Request for Data Collection

The bearer, **Jesse Shema Kyanga** is a student of Adventist University of Central Africa in the Faculty of Information Technology, major in Software Engineering Department and he is working on a dissertation entitled: "**EMPLOYEE PRODUCTIVITY MANAGEMENT SYSTEM**".

The purpose of this letter is to request for your permission to allow him to collect data in your organization.

Any assistance given to him will be highly appreciated.

Yours sincerely,


Mr. ISHIMWE Prince
HOD||Software Engineering
Tel: +250 782 974 477
Email: prince.ishimwe@auca.ac.rw



Approval Letter



12th March 2025

Dear Jesse Shema Kyanga

Subject: ENDORSEMENT OF FINAL YEAR PROJECT CONCEPT

This letter confirms that we have reviewed Shema Jesse's project proposal for an Employee Productivity Management System

MTN Rwanda has approved your request and your project concept, affirming its potential value to our business operations.

Wish you success

Yours faithfully



Iradukunda Willy serge
Supervisor
IT

A handwritten signature in blue ink, appearing to be "Willy", written over the printed name and title.

Sensitivity: MTN Internal